

Android Games Dataset Data Preprocessing, Exploratory Data Analysis & Interactive Dashboard

A structured data-analytics study of 10,000 Android mobile games — covering data cleaning, quality assessment, exploratory analysis, and an interactive dashboard design.



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Course	Master of Computer Applications (MCA)
Project Type	Data Analytics / Data Preprocessing
Dataset	Android Games Dataset (cleaned_data.csv)

Python

Pandas

NumPy

Matplotlib

Seaborn

Plotly

HTML

JavaScript

Chart.js

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01 Abstract

This project presents a complete data-analytics workflow applied to a real-world style **Android Games dataset** containing 10,000 cleaned game records (57 attributes each), sourced from `cleaned_data.csv`, with its pre-cleaning counterpart `android_games_eda_ready.csv` (10,150 records) used as the "before" reference for the preprocessing stage. The objective is to demonstrate, end-to-end, how raw mobile-game analytics data is inspected, cleaned, statistically explored, visualized, and finally packaged into an interactive dashboard.

The preprocessing stage identifies and resolves common data-quality problems found in the raw file, including **150 exact duplicate rows**, missing values across nine columns (such as `multiplayer_support`, `event_theme`, and `marketing_spend_usd`), a small number of **logically invalid values** (ratings above the 5.0 ceiling and negative prices/crash rates), and columns that required type standardisation for numerical analysis. Rather than blindly removing every unusual value, the project distinguishes between values that are *genuinely invalid* (e.g. a rating of 5.98) and values that are *legitimately extreme* (e.g. a viral game with 53 million downloads).

The exploratory data analysis (EDA) stage then investigates the cleaned dataset across multiple dimensions: game **genres and sub-genres**, **rating** distribution and quality, **review** and **download** behaviour, **monetization model** (Free / Freemium / Paid / Subscription), **content/age rating** composition, and the statistical **correlations** between engagement, retention and revenue metrics. Over 20 professional visualizations — bar charts, histograms, box plots, pie/doughnut charts, scatter plots, line-trend charts, and a correlation heatmap — are produced using Matplotlib and Seaborn.

Finally, the project outlines the design of an **interactive analytics dashboard** (HTML, JavaScript and Chart.js) that lets a user filter games by category, monetization type, content rating, genre, and minimum rating, and instantly view KPI cards and charts that update accordingly. The report concludes with key insights drawn strictly from the dataset and recommendations for future enhancement, such as machine-learning-based rating and popularity prediction.

02 Introduction

2.1 What is Data Preprocessing?

Data preprocessing is the set of techniques used to transform raw, imperfect data into a clean, consistent, and analysis-ready format. Real-world datasets — including mobile-app analytics exports — commonly contain missing fields, duplicate records, inconsistent categories, incorrect data types, and outlying or invalid values. Preprocessing detects and corrects these issues before any statistical or visual analysis is performed.

2.2 Why Preprocessing is Important

Without preprocessing, analytical results can be biased or outright misleading. Duplicate rows inflate counts and averages; missing values break aggregate calculations; and invalid values (such as a rating outside the valid 0–5 scale) distort statistics like the mean and standard deviation. Clean data is the foundation of every trustworthy analysis and dashboard.

2.3 What is Exploratory Data Analysis (EDA)?

Exploratory Data Analysis is the process of summarizing a dataset's main characteristics, often using visual and statistical methods, before drawing conclusions or building models. EDA answers questions such as: What does the distribution of ratings look like? Which genres dominate the market? Do more reviews correlate with higher ratings?

2.4 Why Visualization is Useful

Charts communicate patterns far faster than raw numbers. A histogram instantly reveals whether ratings cluster around a value; a scatter plot reveals whether two variables move together; a heatmap condenses dozens of correlation values into one glanceable grid.

2.5 Why Android Game Data is Useful for Analysis

Mobile game data combines behavioural metrics (retention, session length), business metrics (revenue, ARPU, marketing spend), and catalog metadata (genre, monetization model, content rating). This makes it an excellent case study for practising the full data-analytics pipeline — from cleaning to dashboarding — on a realistically messy, multi-dimensional dataset.

2.6 Project Workflow



03 Dataset Description

The analysis uses two versions of the same dataset: **android_games_eda_ready.csv** — the raw, pre-cleaning export (10,150 rows) — and **cleaned_data.csv** — the deduplicated, analysis-ready dataset (10,000 rows). Both share the identical 57-column schema. All EDA in this report is performed on `cleaned_data.csv`.

10,000 ROWS (CLEANED)	57 COLUMNS	29 NUMERICAL COLUMNS	28 CATEGORICAL / BOOLEAN COLUMNS
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Column-type split: 29 numeric (int64 / float64), 5 boolean, and 23 text/categorical columns (including date fields stored as strings and identifier fields).

3.1 Column Reference Table

Column Name	Data Type	Description	Example	Missing (cleaned)
game_id	int64	Unique numeric identifier for each game	622	0
game_name	text	Name of the game	Game_622_Arena	0
developer_name	text	Studio / developer that published the game	Studio_205	0
package_name	text	Android package identifier	com.studio1664.game622	0
genre	text (categorical)	Primary game category	Casual	0
sub_genre	text (categorical)	Specific sub-genre / niche	Party Game	0
monetization_model	text (categorical)	Revenue model: Free, Freemium, Paid, Subscription	Freemium	0
contains_ads	text (Yes/No)	Whether the game shows in-app ads	Yes	0
has_in_app_purchases	text (Yes/No)	Whether the game offers in-app purchases	Yes	0
multiplayer_support	text (categorical)	Multiplayer mode: Online, Local, Both	Online	4,013
age_rating	text (categorical)	Content/age rating classification	Mature 17+	0
art_style	text (categorical)	Visual art style of the game	2D Cartoon	0
engine_used	text (categorical)	Game engine used for development	Unity	0
region_focus	text (categorical)	Primary target market/region	Global	0
publisher_tier	text (categorical)	Size classification of publisher	Indie	0
event_theme	text (categorical)	Seasonal/marketing event theme, if any	Summer Fest	7,577
release_date	date (string)	Public release date	2025-05-16	0
soft_launch_date	date (string)	Soft-launch date, if applicable	2025-04-06	1,000
last_update_date	date (string)	Date of most recent update	2026-04-08	0
was_featured_on_store	bool	Whether featured on the Play Store	False	0
featured_start_date	date (string)	Start date of store feature (if featured)	2019-01-31	6,697
featured_end_date	date (string)	End date of store feature (if featured)	2019-02-07	6,697
price_usd	float64	Listed price in USD	0.0	0
downloads	int64	Total number of downloads / installs	70,113	0
retention_day1_pct	float64	Percentage of users retained after day 1	31.85	0
retention_day7_pct	float64	Percentage of users retained after day 7	15.09	0
retention_day30_pct	float64	Percentage of users retained after day 30	4.29	0
active_users_30d	int64	Active users in the trailing 30 days	3,511	0
avg_session_minutes	float64	Average session duration in minutes	14.91	0

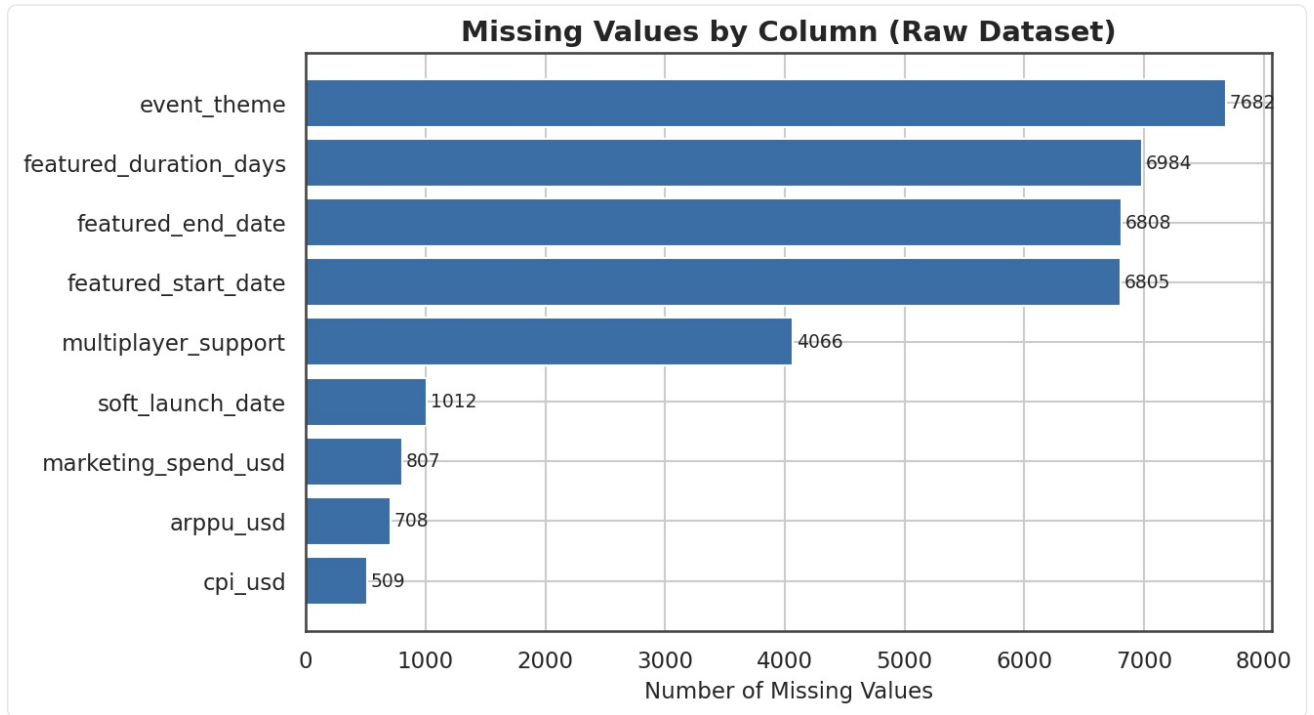
avg_daily_sessions	float64	Average number of sessions per day	1.73	0
crash_rate_pct	float64	Percentage of sessions ending in a crash	2.04	0
apk_size_mb	float64	Application package size in MB	244.04	0
update_frequency_days	float64	Average days between updates	62.0	0
rating_avg	float64	Average user star rating	3.48	0
rating_count	int64	Total number of ratings received	7,295	0
review_count	int64	Total number of written reviews	3,095	0
marketing_spend_usd	float64	Marketing budget spent (USD)	4,851.05	800
cpi_usd	float64	Cost-per-install (USD)	1.08	500
conversion_to_payer_pct	float64	% of users who convert to paying users	5.96	0
ad_impressions_per_user	float64	Average ad impressions per user	28.4	0
ad_revenue_share_pct	float64	% of revenue from advertising	35.35	0
iap_revenue_share_pct	float64	% of revenue from in-app purchases	69.4	0
arpu_usd	float64	Average revenue per user (USD)	0.94	0
arppu_usd	float64	Average revenue per paying user (USD)	3.66	700
wishlist_prelaunch	int64	Pre-launch wishlist count	644	0
total_revenue_usd	float64	Total revenue generated (USD)	15,593.31	0
is_weekend_release	bool	Whether released on a weekend	False	0
has_seasonal_event	bool	Whether the game runs seasonal events	False	0
cross_platform_available	bool	Cross-platform availability	True	0
cloud_save_support	bool	Cloud-save feature support	True	0
is_hit_game	int64 (flag 0/1)	Flag marking breakout "hit" games	0	0
post_30d_revenue_usd	float64	Revenue generated after the first 30 days	9,963.75	0
post_30d_rating_count	int64	Rating count after the first 30 days	6,328	0
featured_duration_days	float64	Number of days the game was featured	7.0	6,871
store_platform	text (constant)	Store platform (always "Android")	Android	0
country_code_primary	text (categorical)	Primary country code of users	IN	0
row_checksum_id	text	Internal checksum / row identifier	CHK_100621	0

Note: several "missing" columns are *structurally* missing rather than data-entry errors — e.g. featured_start_date is empty for the 6,697 games where was_featured_on_store is False, and featured_duration_days is only populated for games that were actually featured.

04 Data Quality Analysis

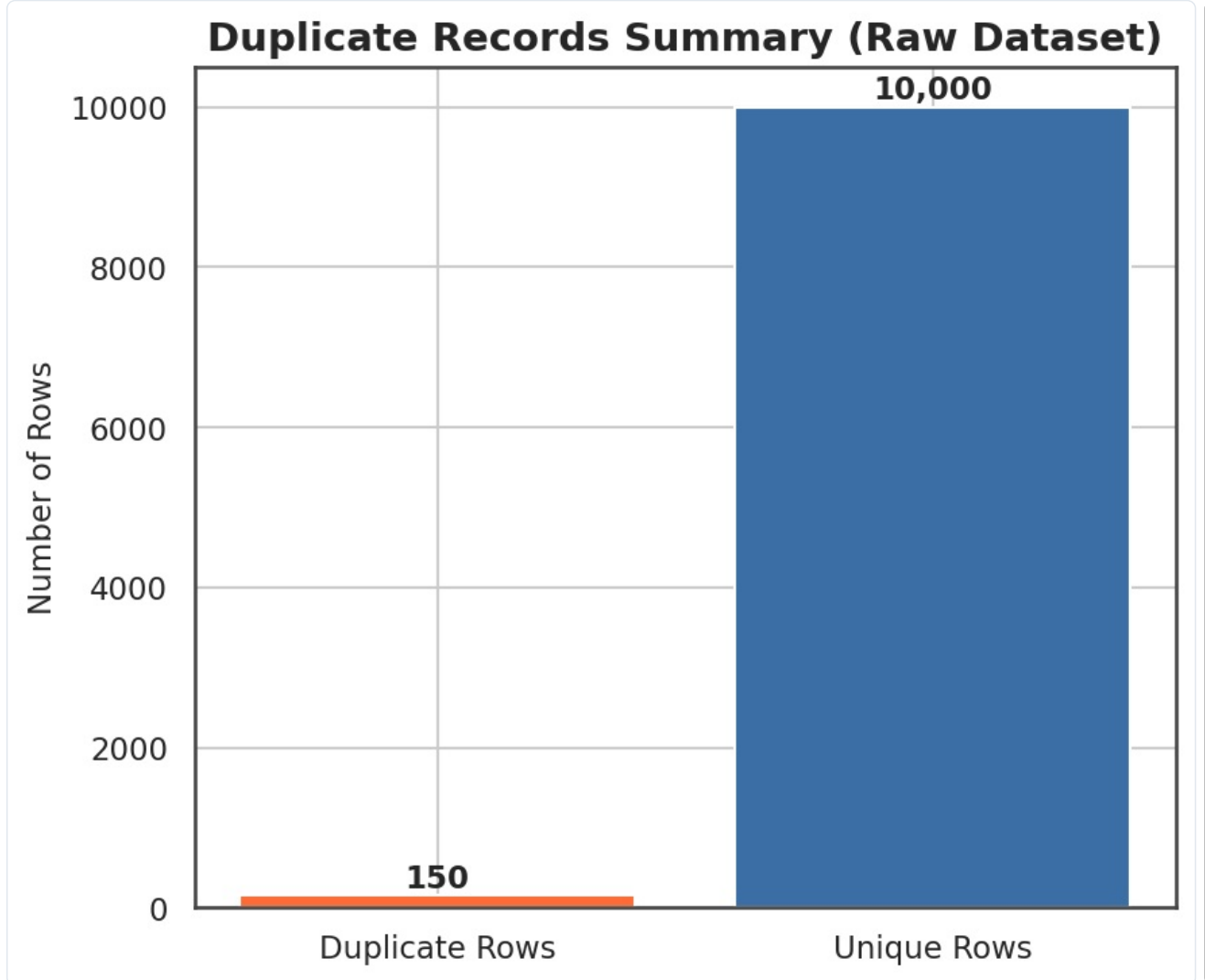
Before cleaning, the raw dataset (`android_games_eda_ready.csv`, 10,150 rows) was assessed for missing values, duplicate rows, invalid values, and outliers.

4.1 Missing Values



Nine of the 57 columns contain missing values. The largest gaps are in `featured_duration_days` (6,984 missing), `event_theme` (7,682 missing), `featured_start_date/featured_end_date` (~6,800 missing), and `multiplayer_support` (4,066 missing) — all of which are structurally missing rather than erroneous, since they only apply to games with a specific attribute (e.g. only "featured" games have feature dates).

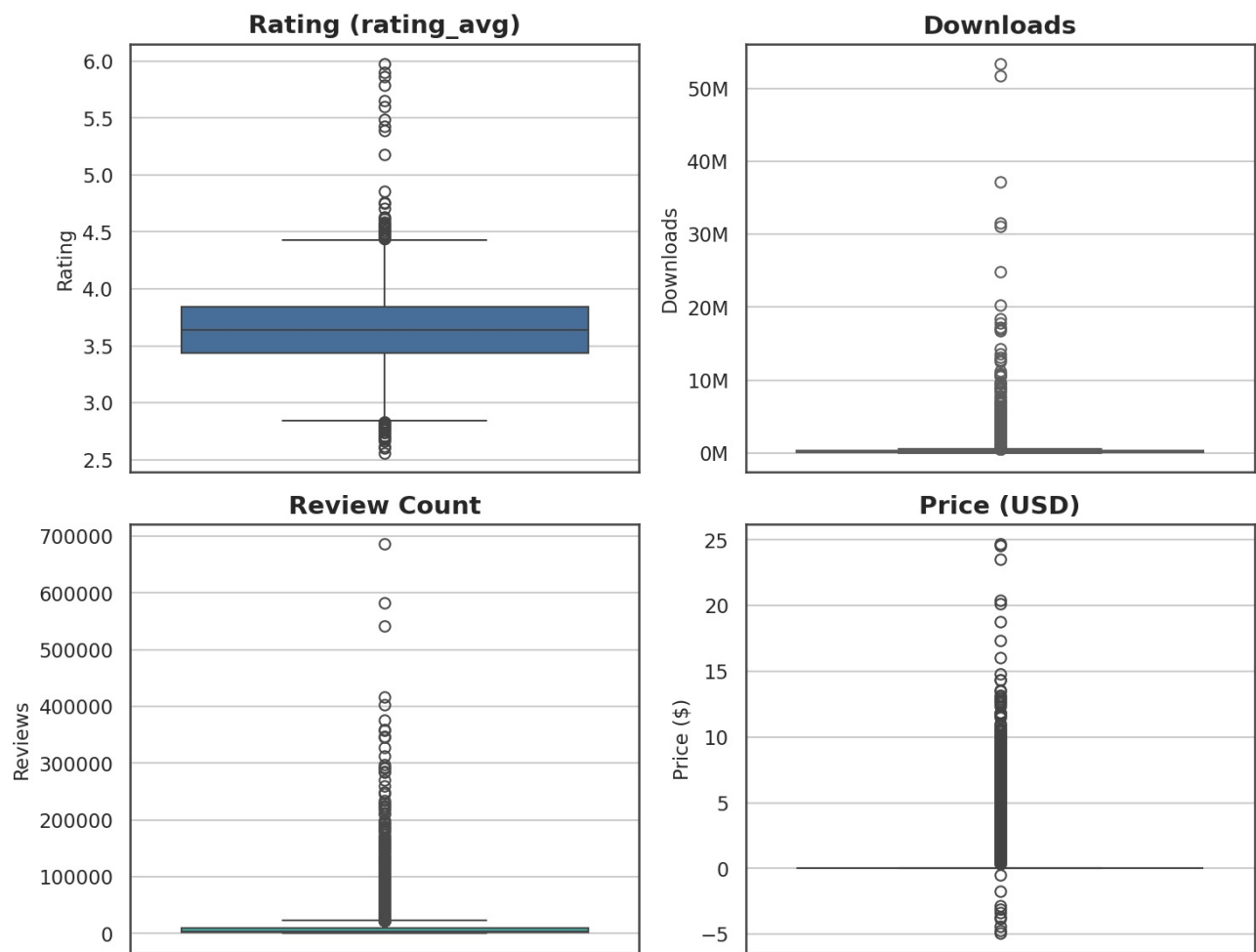
4.2 Duplicate Records



The raw dataset contained **150 fully duplicated rows** (exact duplicates across every column, including `game_id`) out of 10,150 total rows — a duplication rate of 1.48%. These were removed during cleaning.

4.3 Outlier Detection (Boxplots)

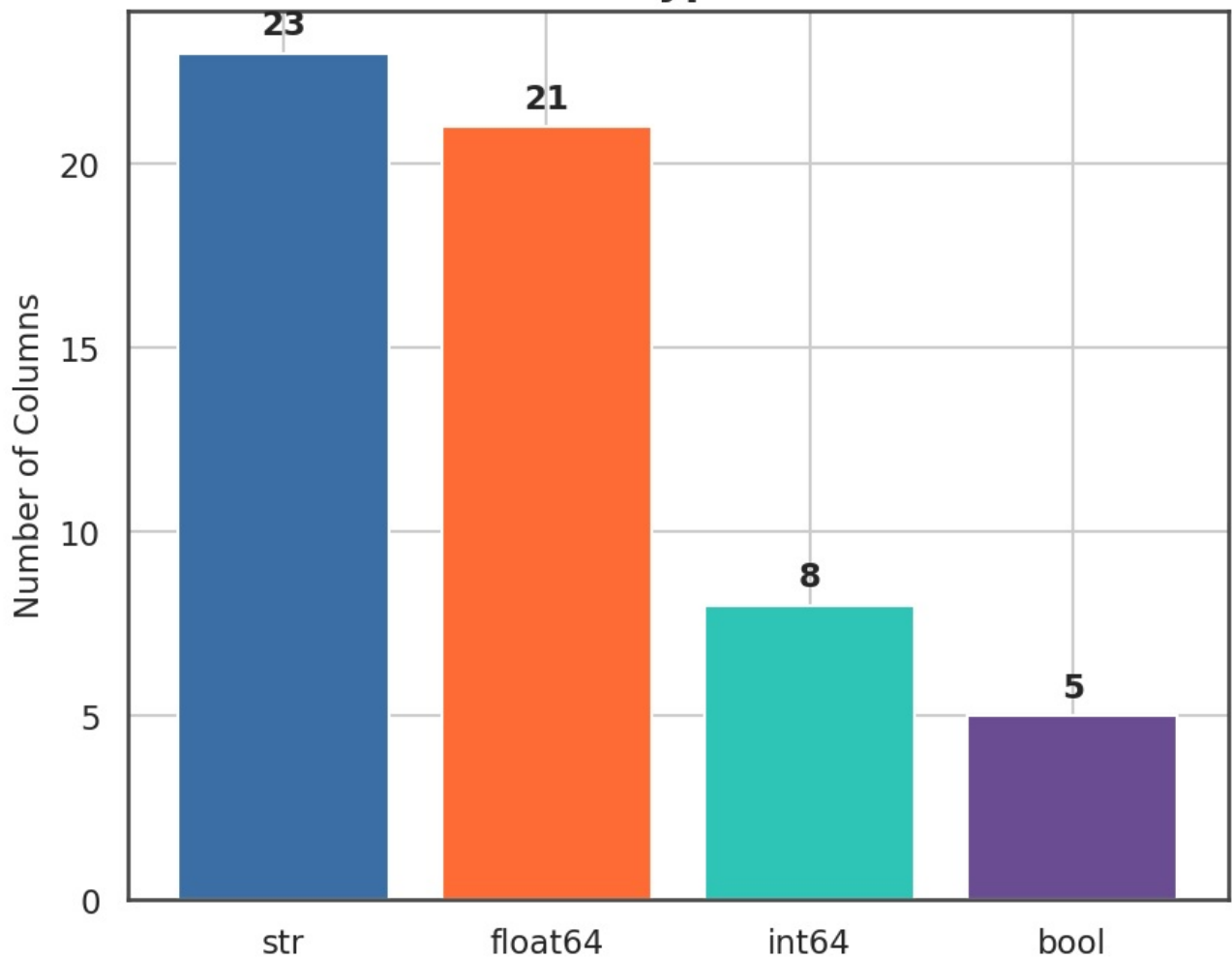
Outlier Detection - Boxplots of Key Numerical Columns



Boxplots of `rating_avg`, `downloads`, `review_count`, and `price_usd` show a right-skewed distribution for downloads and reviews (typical for app-store data, where a small number of viral games dominate), while rating and price show a tighter spread with a handful of extreme points.

4.4 Data Type Distribution

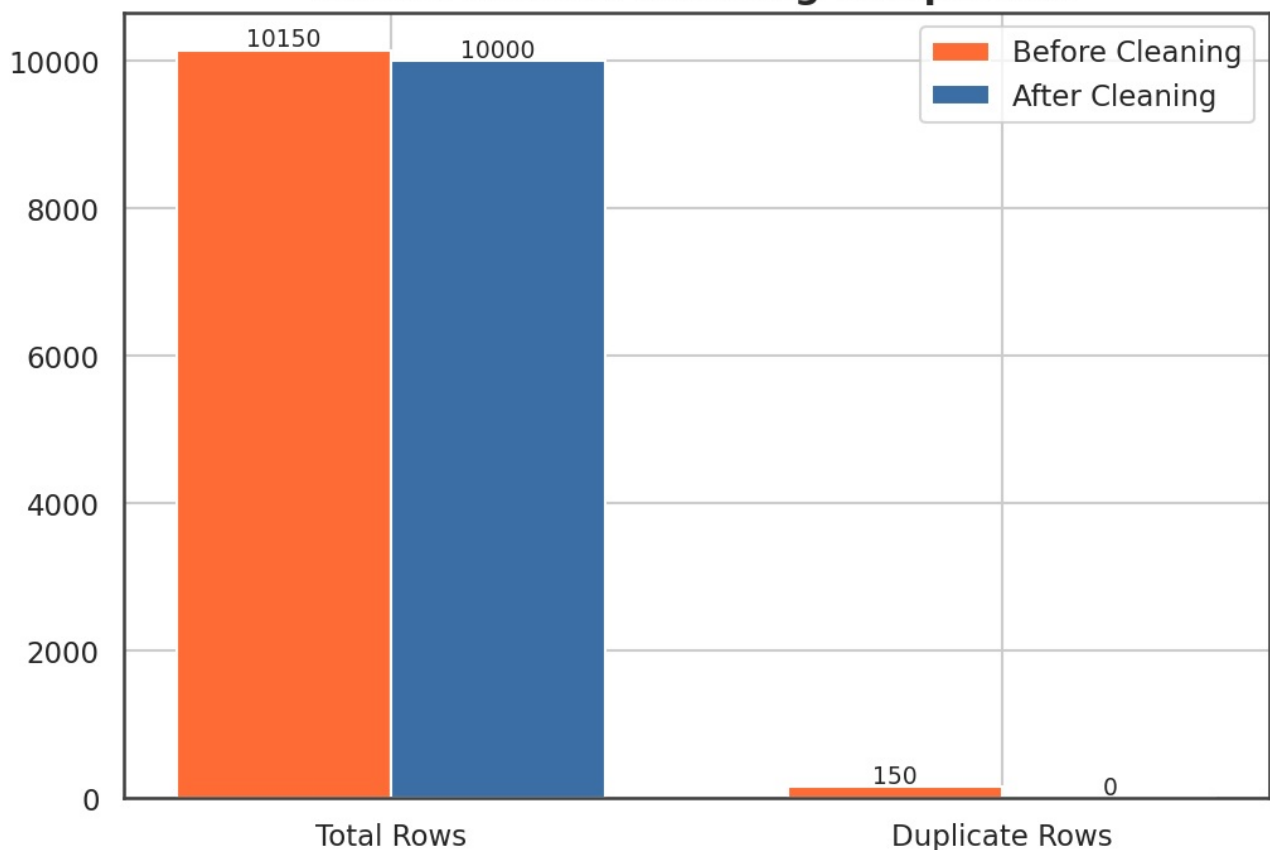
Column Data Type Distribution



The raw file loads with 29 numeric columns, 5 boolean columns, and 23 text/object columns — matching the cleaned dataset, confirming that no column needed a structural type change during cleaning (values were already numeric where expected).

4.5 Before vs After Cleaning

Before vs After Cleaning Comparison



4.6 Invalid Value Findings

Logically invalid values identified in the dataset:

- **10 games** have rating_avg above the valid 5.0 ceiling (max observed: **5.98**) — these are data-entry anomalies rather than true ratings.
- **10 games** have a negative price_usd (as low as **−\$4.94**), which is not possible for a real listing.
- **10 games** have a negative crash_rate_pct (minimum **−2.78%**), which is not physically meaningful for a percentage metric.

These invalid values are flagged and excluded from range-sensitive calculations (e.g. rating averages reported in Section 9 exclude ratings above 5.0) rather than being silently imputed, to avoid fabricating data.

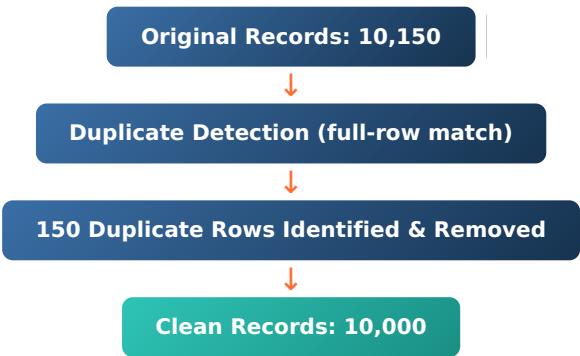
05 Data Preprocessing

5.1 Handling Missing Values

The nine columns with missing values were reviewed individually. Because each gap corresponds to a real absence of the underlying event (e.g. a game that was never featured has no feature dates), these fields were **left as missing (NaN)** rather than imputed with fabricated values, which would distort downstream statistics. Aggregations in this report (means, medians, correlations) are computed only over non-missing values for the relevant column.

Column	Before (raw)	Cleaning Action	After (cleaned)
multiplayer_support	4,066 missing	Retained as NaN — represents "no multiplayer data recorded"	4,013 missing
event_theme	7,682 missing	Retained as NaN — game has no seasonal event	7,577 missing
featured_start_date / featured_end_date	~6,805 missing each	Retained as NaN — game was never featured	~6,697 missing each
featured_duration_days	6,984 missing	Retained as NaN — only defined for featured games	6,871 missing
marketing_spend_usd	807 missing	Retained as NaN — spend not tracked for this record	800 missing
cpi_usd	509 missing	Retained as NaN — no install-cost data	500 missing
arppu_usd	708 missing	Retained as NaN — no paying users to compute average from	700 missing
soft_launch_date	1,012 missing	Retained as NaN — game skipped a soft launch phase	1,000 missing

5.2 Duplicate Removal



5.3 Outlier Detection

The Interquartile Range (IQR) method ($Q1 - 1.5 \times IQR$ to $Q3 + 1.5 \times IQR$) was applied to key numerical columns:

Column	Q1	Q3	Upper Bound	Values Beyond Upper Bound
downloads	46,458	267,941	600,166	995 games (9.95%)
review_count	1,564	10,100	22,903	1,067 games (10.67%)
rating_avg	3.44	3.84	4.44 (and lower bound 2.84)	96 games (0.96%)

Boxplots for these columns are shown in Section 4.3 and Section 15. Because downloads and reviews are naturally right-skewed (a small share of hit games account for a large share of installs), most "IQR outliers" here represent genuinely high-performing games rather than data errors — see Section 15 for further discussion.

5.4 Data Type Conversion

All numerical fields relevant to this analysis were already stored in correct numeric form in the source CSV (int64/float64), so no explicit type coercion was required for `rating_avg`, `review_count`, `downloads`, or `price_usd`. Date columns (`release_date`, `last_update_date`, etc.) are stored as text strings and were parsed to datetime objects only where a time-based trend (Section 6, Section 14) required it.

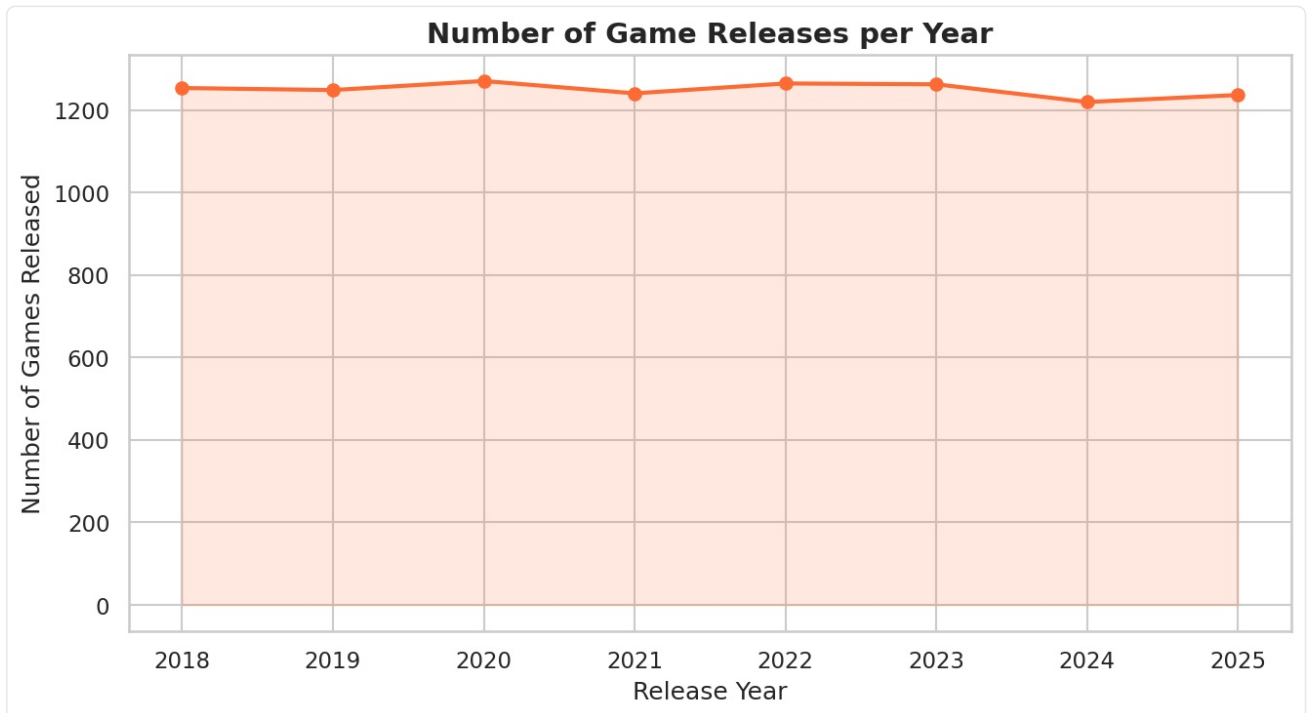
5.5 Categorical Data Cleaning

Categorical fields such as `genre` (8 unique values), `monetization_model` (4 unique values), and `age_rating` (3 unique values) were checked for inconsistent casing, whitespace, or duplicate-meaning labels (e.g. "Casual" vs "casual"). No inconsistent category spellings were found in the cleaned file — all categorical columns already use a single, consistent label per category.

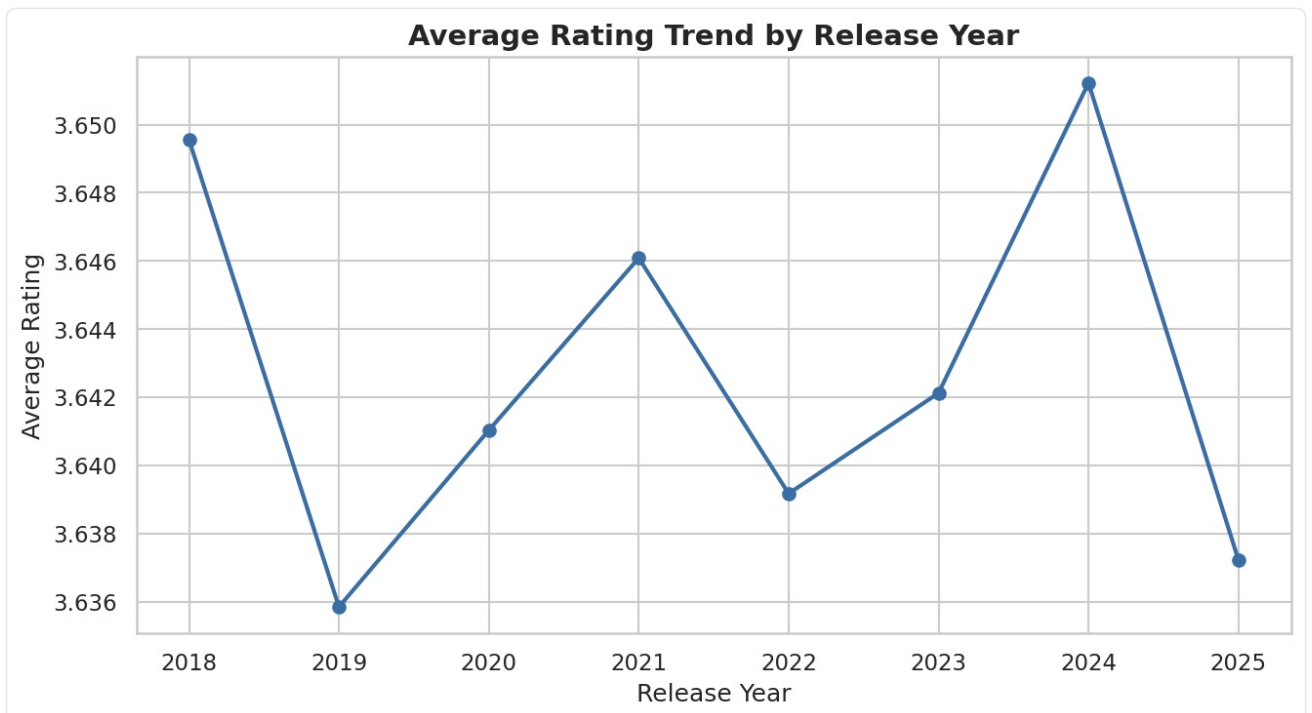
06 Exploratory Data Analysis

With the dataset cleaned, this section (and Sections 7–15) explores the data using a mix of chart types — bar, line, pie/doughnut, histogram, box plot, scatter plot, and heatmap — to surface patterns across genres, ratings, reviews, installs, monetization, and correlations.

6.1 Release Trend Over Time (Line Charts)



The number of game releases in the dataset per year, based on `release_date` (years with 10+ releases shown). This line chart shows how the sampled catalog's release volume has moved over time.

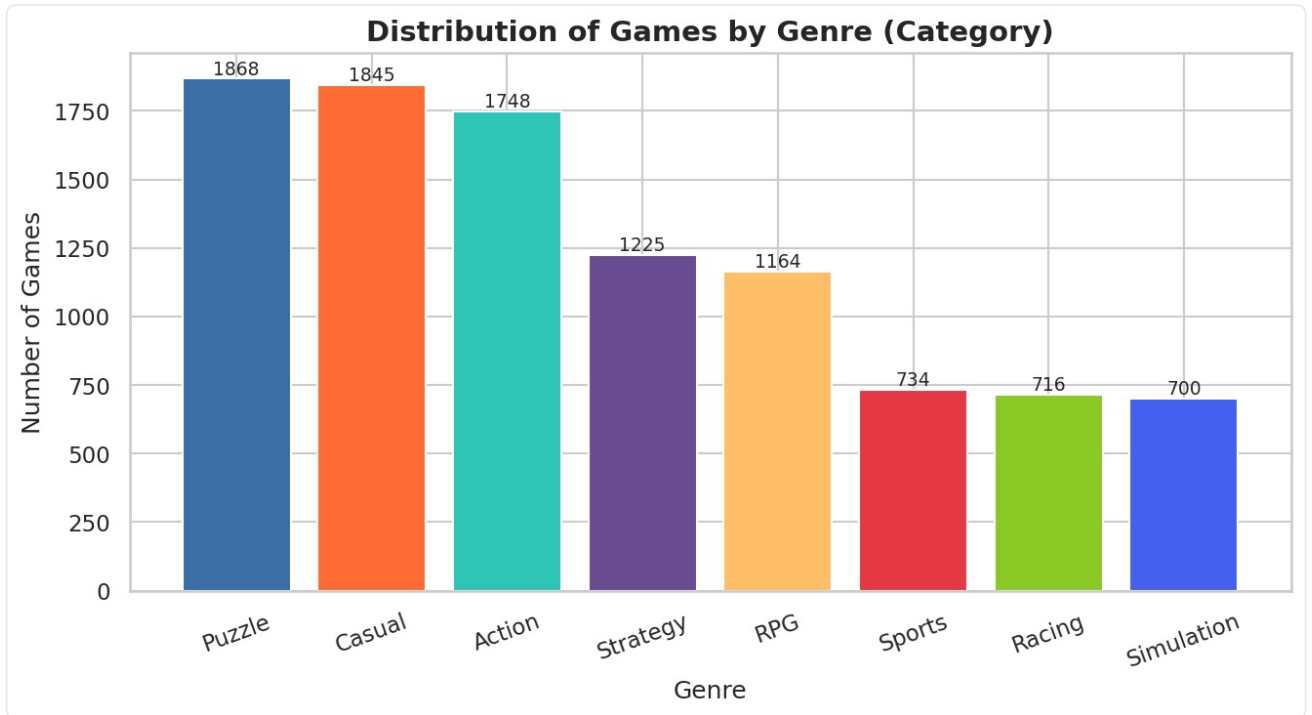


Average `rating_avg` by release year. Ratings stay within a narrow band across years, suggesting release year alone is not a strong driver of average rating in this dataset.

07 Category (Genre) Analysis

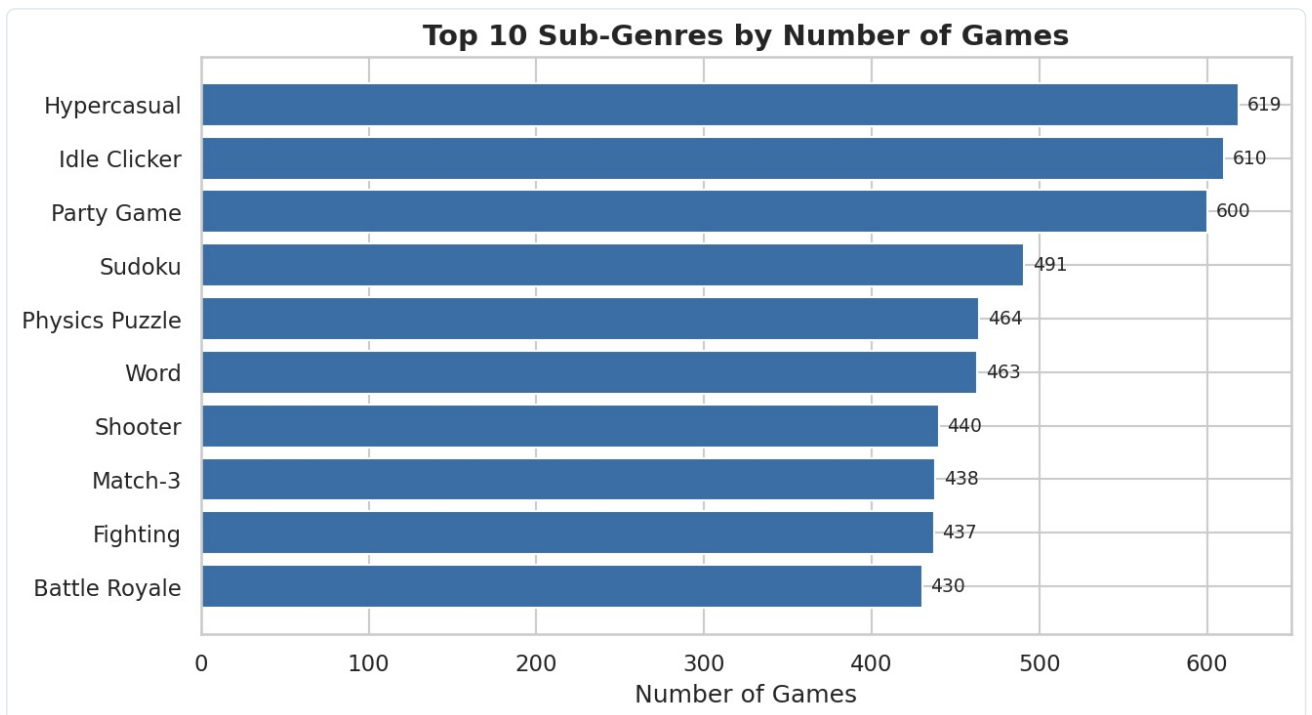
The dataset's genre column serves as the primary "category" field (8 categories), with sub_genre providing a finer-grained breakdown (31 unique values).

7.1 Category Distribution



Puzzle (1,868 games) and Casual (1,845 games) are the two largest categories, followed by Action (1,748). Racing (716) and Simulation (700) are the smallest categories in the dataset.

7.2 Top Sub-Genres



Hypercasual (619), Idle Clicker (610), and Party Game (600) are the most common sub-genres, reflecting a dataset weighted toward lightweight, high-volume casual game formats.

Key finding: Puzzle is the most common primary genre (1,868 games, 18.7% of the catalog), while Hypercasual is the single most common sub-genre (619 games).

08 Rating Analysis

3.64

AVERAGE RATING

2.56

MINIMUM RATING

5.98*

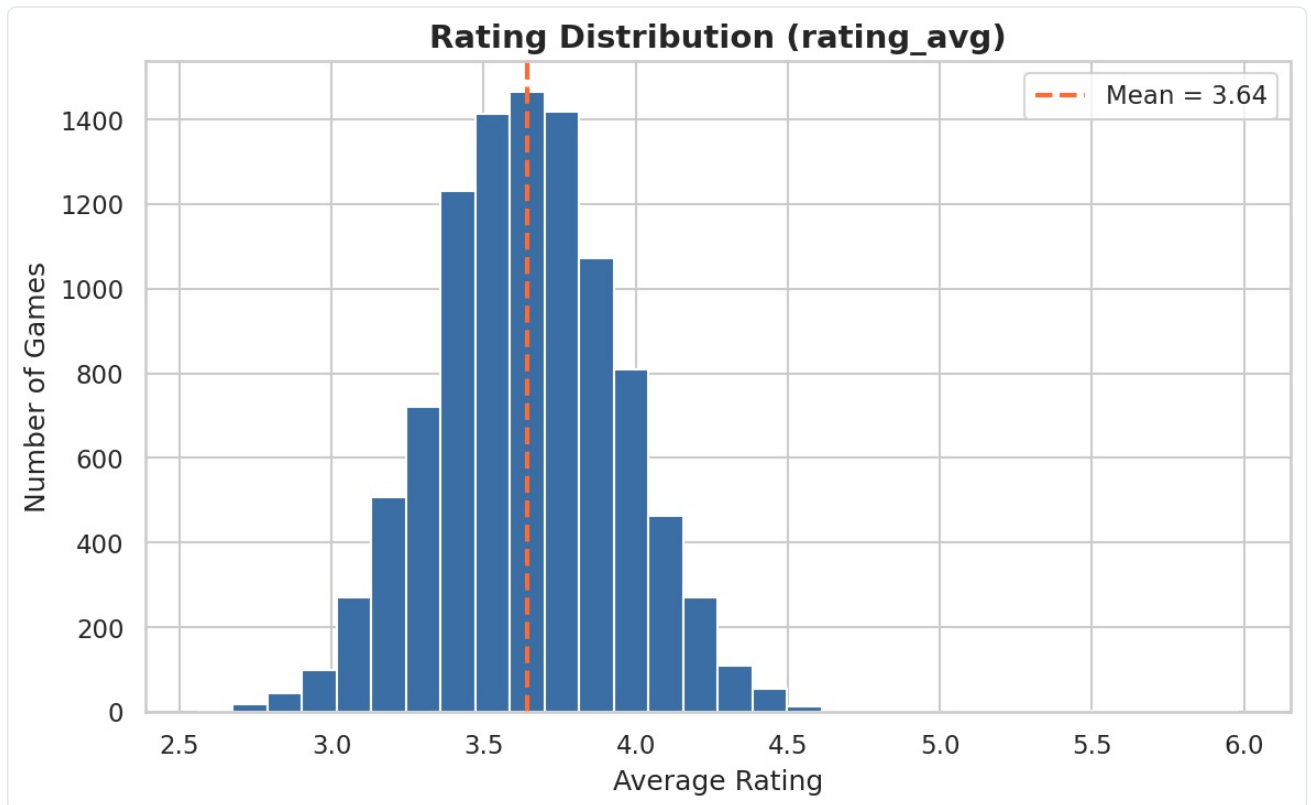
MAXIMUM RATING

3.64

MEDIAN RATING

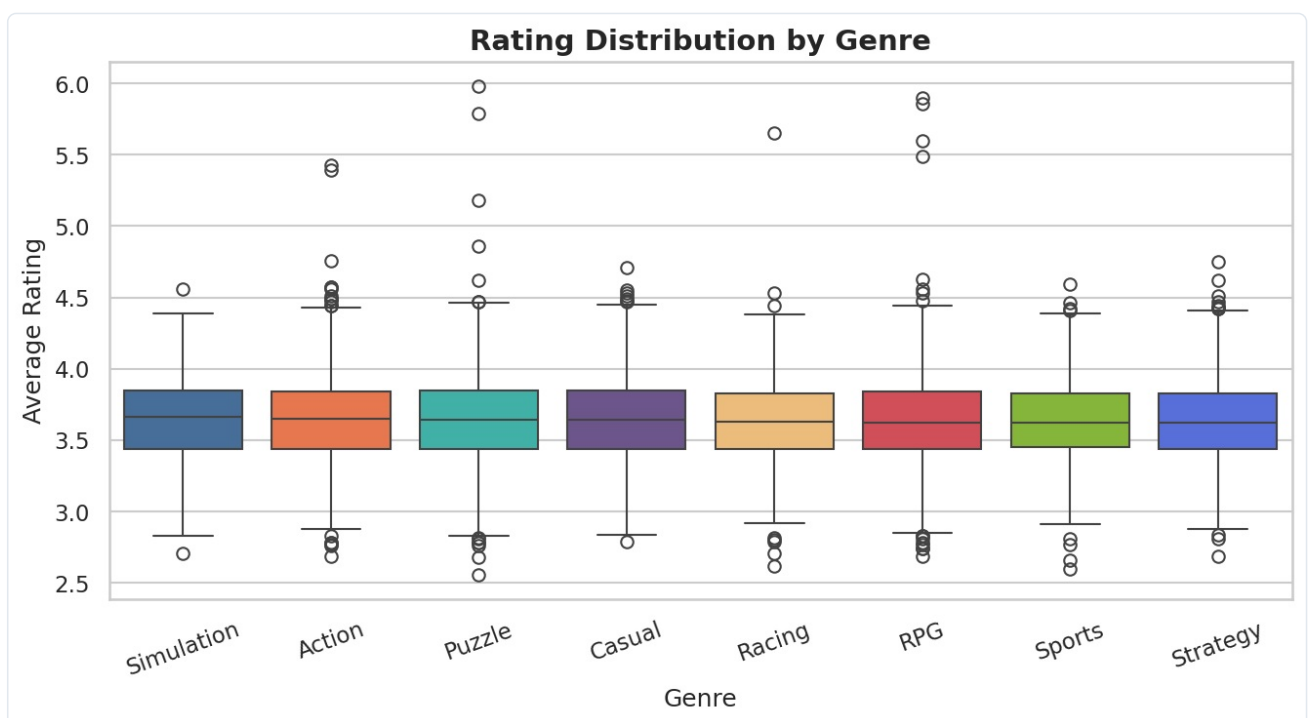
*The maximum of 5.98 falls outside the valid 0–5 rating scale and is one of the 10 invalid values identified in Section 4.6. Excluding the 10 invalid values above 5.0, the maximum valid rating in the dataset is **4.99**.

8.1 Rating Distribution



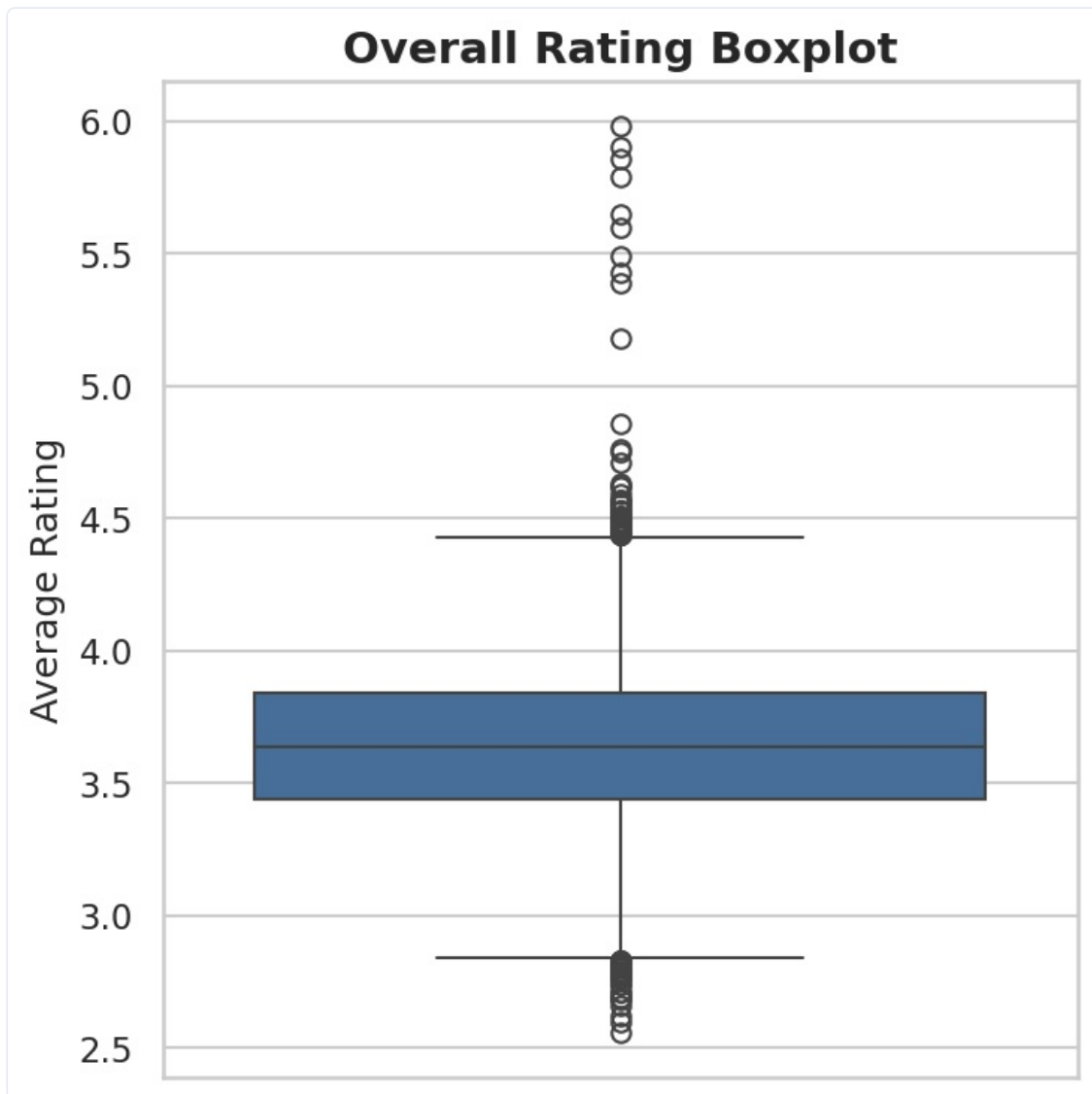
Ratings are approximately normally distributed and tightly clustered around the mean of 3.64, with a standard deviation of 0.31 — most games sit between roughly 3.4 and 3.9 stars.

8.2 Rating by Genre



Median ratings are very similar across all eight genres (roughly 3.6–3.7), indicating that genre choice alone has minimal impact on user rating in this dataset.

8.3 Overall Rating Boxplot



The boxplot confirms a tight interquartile range (3.44–3.84) with a small number of outlying points above 4.4 and the ten invalid points above 5.0 visible at the top.

09 Reviews Analysis

10,854

AVG. REVIEW COUNT

4,029

MEDIAN REVIEW COUNT

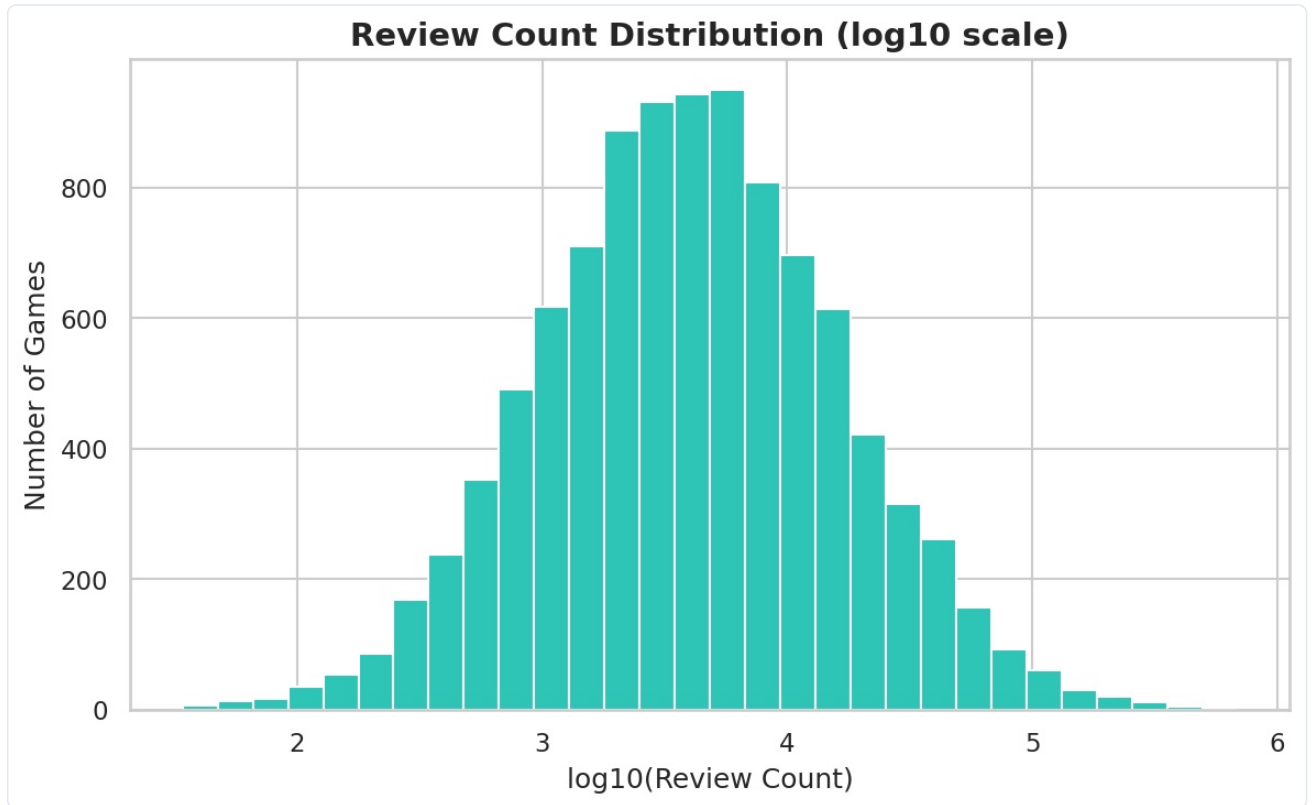
686,581

MAX REVIEW COUNT

108.5M

TOTAL REVIEWS (ALL GAMES)

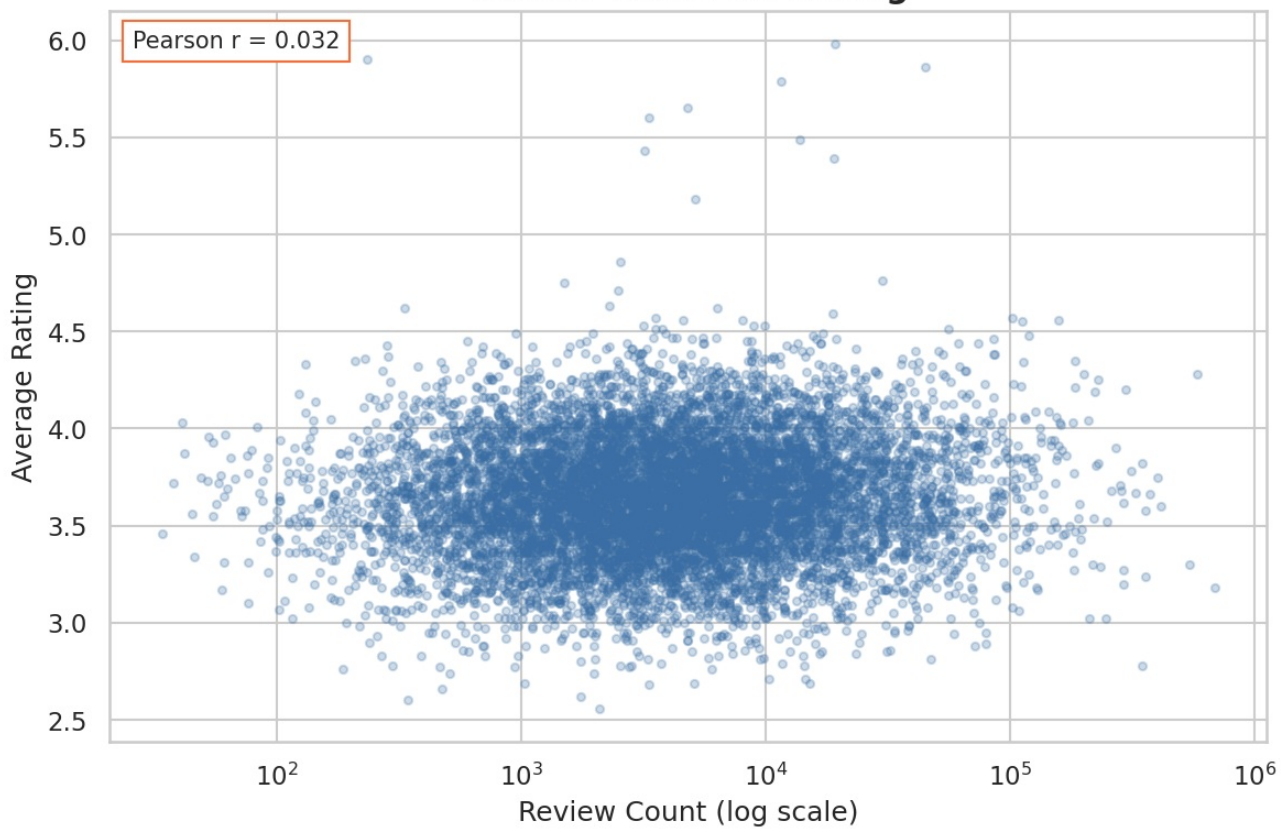
9.1 Reviews Distribution



Review counts are heavily right-skewed (shown here on a log10 scale for readability): most games receive a modest number of reviews, while a small set of breakout titles accumulate hundreds of thousands.

9.2 Reviews vs. Rating

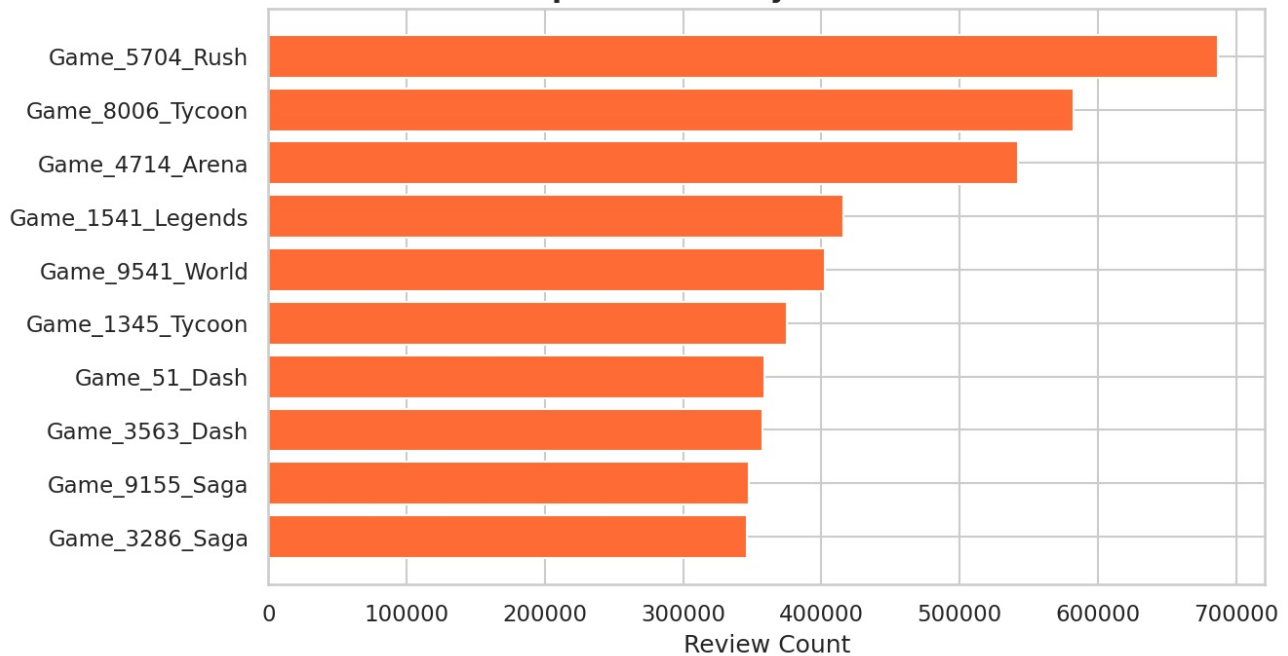
Review Count vs. Rating



The Pearson correlation between `review_count` and `rating_avg` is $r = 0.032$ — a very weak positive relationship. In other words, games with more reviews are **not** meaningfully more (or less) highly rated than games with fewer reviews; review volume and review quality appear to be largely independent in this dataset.

9.3 Top Applications by Reviews

Top 10 Games by Review Count



"Game_5704_Rush" (Sports genre, developer Studio_1395) leads with 686,581 reviews, followed by the other top-10 titles shown above.

10 Installs (Downloads) Analysis

The dataset does not contain a Google-Play-style bucketed "Installs" column (e.g. "1,000,000+"); instead it provides an exact downloads count per game, which is used as the install metric throughout this report.

320,575

AVG. DOWNLOADS

110,610

MEDIAN DOWNLOADS

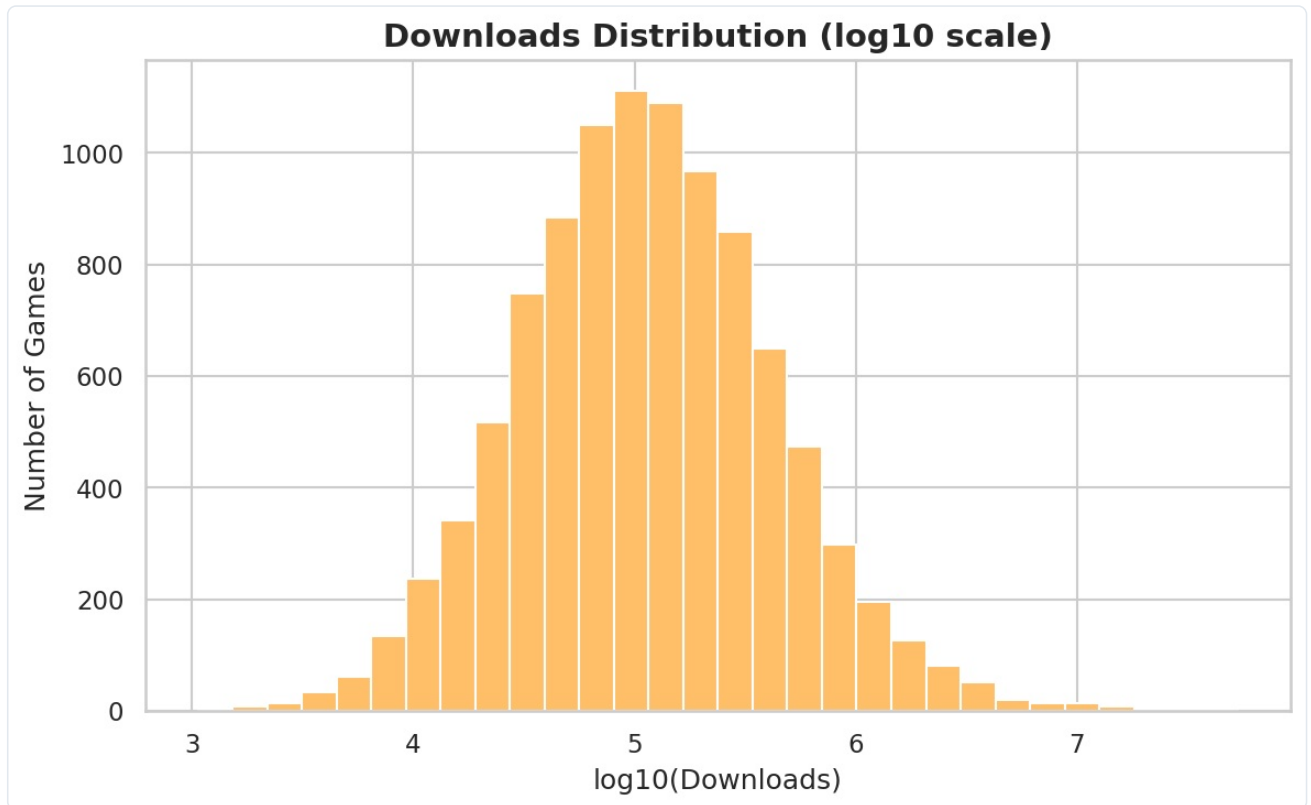
53.45M

MAX DOWNLOADS (SINGLE GAME)

3.21B

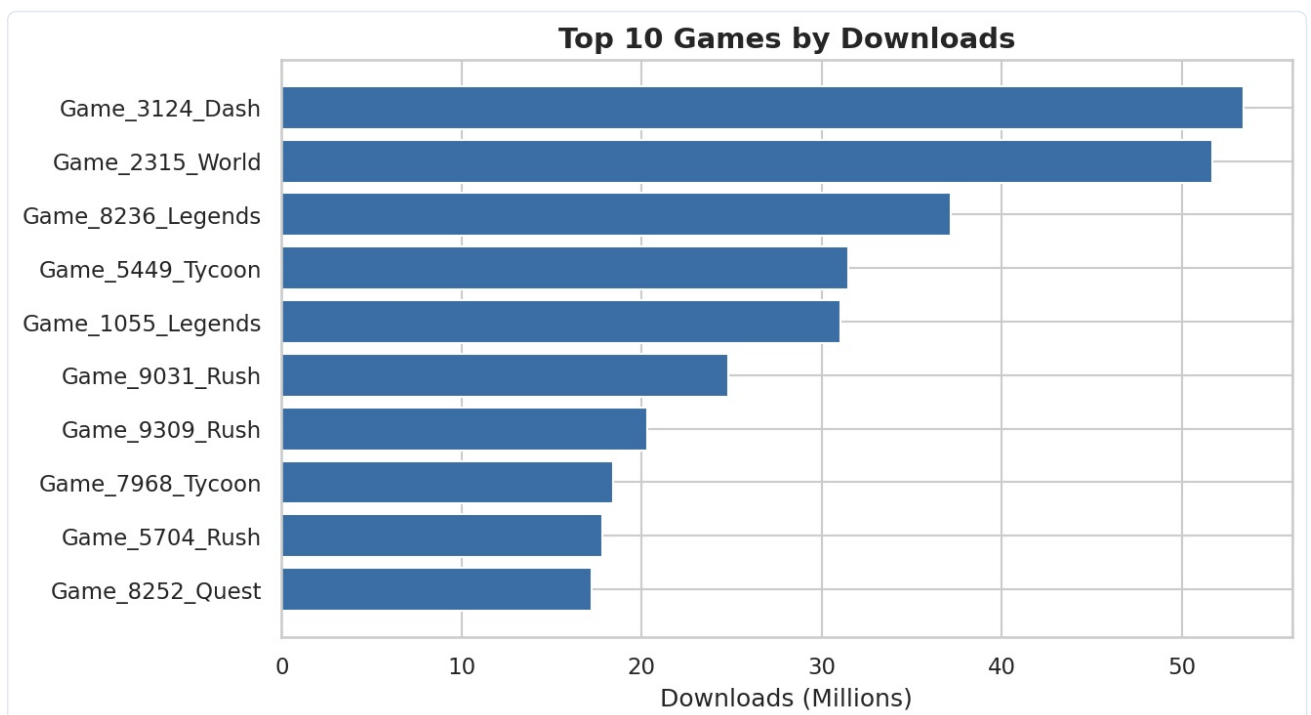
TOTAL DOWNLOADS (ALL GAMES)

10.1 Downloads Distribution



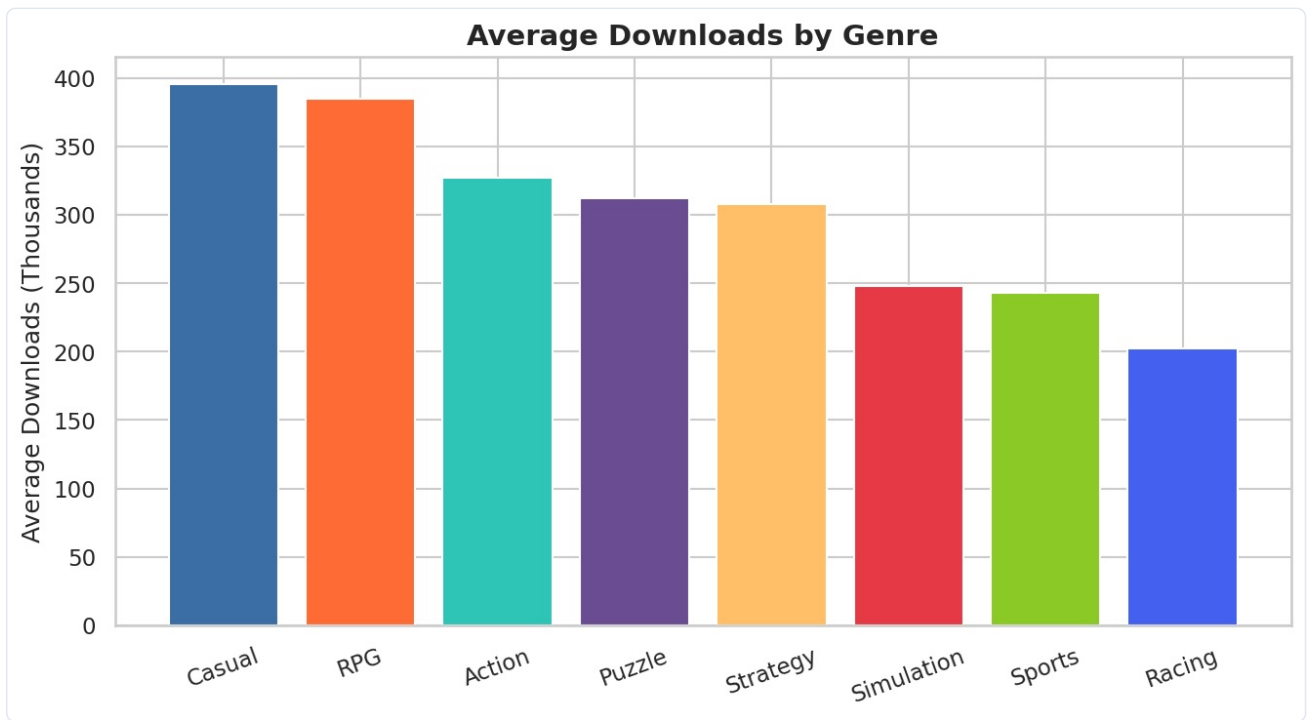
Like reviews, downloads follow a strongly right-skewed, power-law-like distribution (shown on a log10 scale), typical of app marketplaces where a small share of titles capture most installs.

10.2 Top Installed Games



"Game_3124_Dash" (RPG genre, developer Studio_2080) is the most-downloaded game in the dataset with 53.45 million downloads.

10.3 Average Downloads by Genre



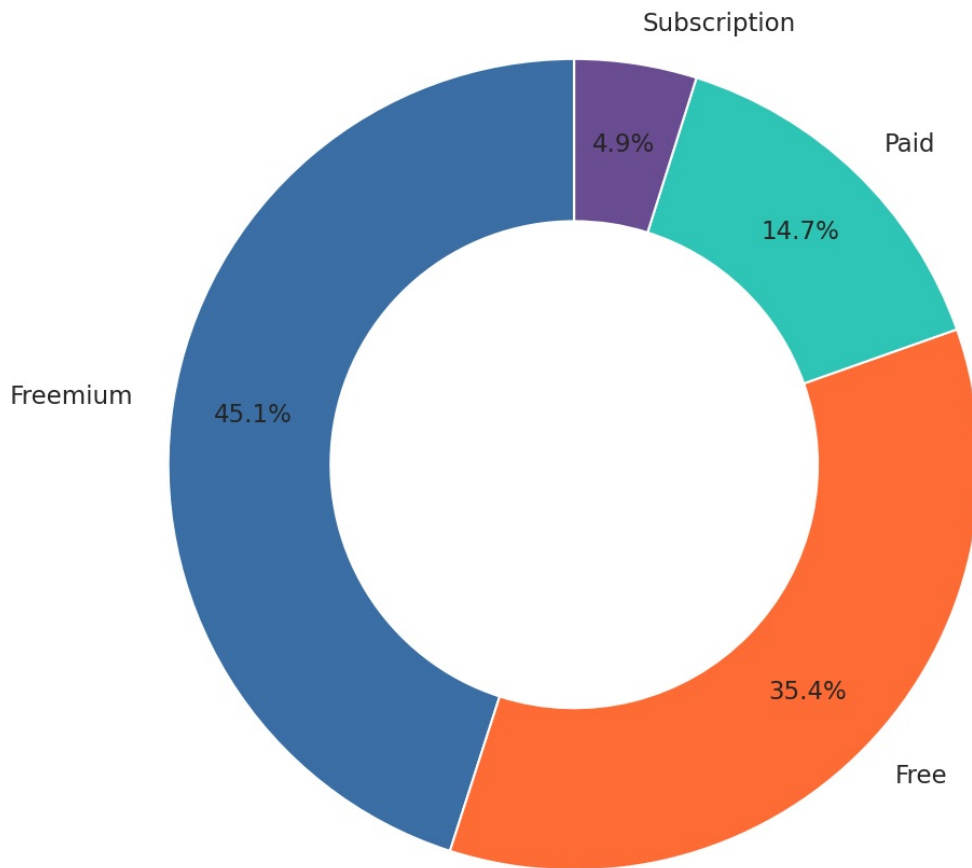
Casual and RPG genres show the highest average downloads per game, while Racing shows the lowest, though this is influenced by a small number of very large outlier titles within each genre.

11 Monetization Model Analysis (Free vs Paid)

The dataset does not use a simple binary "Type" column; instead `monetization_model` captures four categories: **Free**, **Freemium**, **Paid**, and **Subscription**. A simpler price-based split is also available via `price_usd` (0 = free listing, >0 = paid listing).

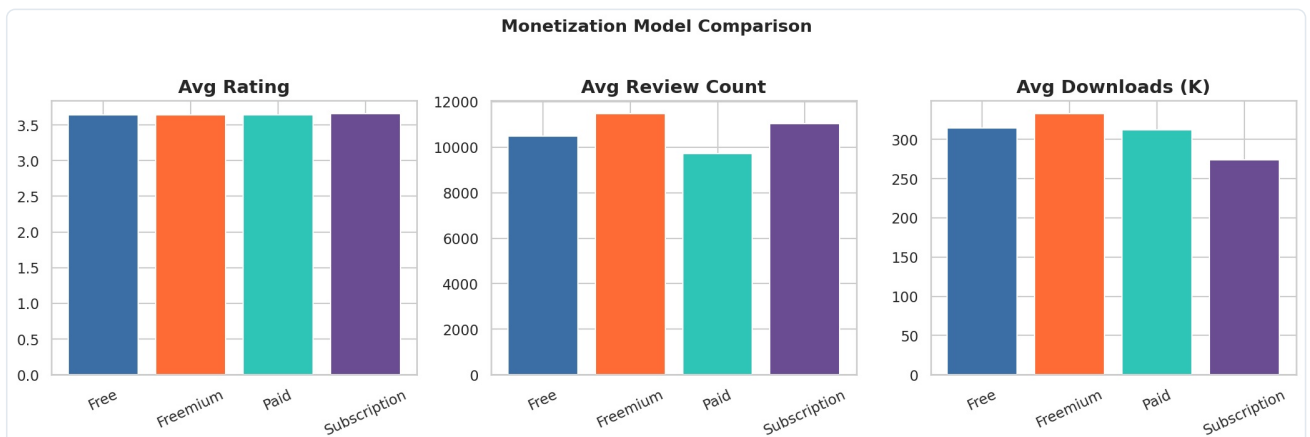
11.1 Distribution

Monetization Model Distribution (Free / Freemium / Paid / Subscription)



Freemium is the leading model (4,505 games, 45.1%), followed by Free (3,536 games, 35.4%), Paid (1,472 games, 14.7%), and Subscription (487 games, 4.9%). By simple listing price, 8,034 games (80.3%) are free to download and 1,956 (19.6%) carry a positive price.

11.2 Comparison Across Models



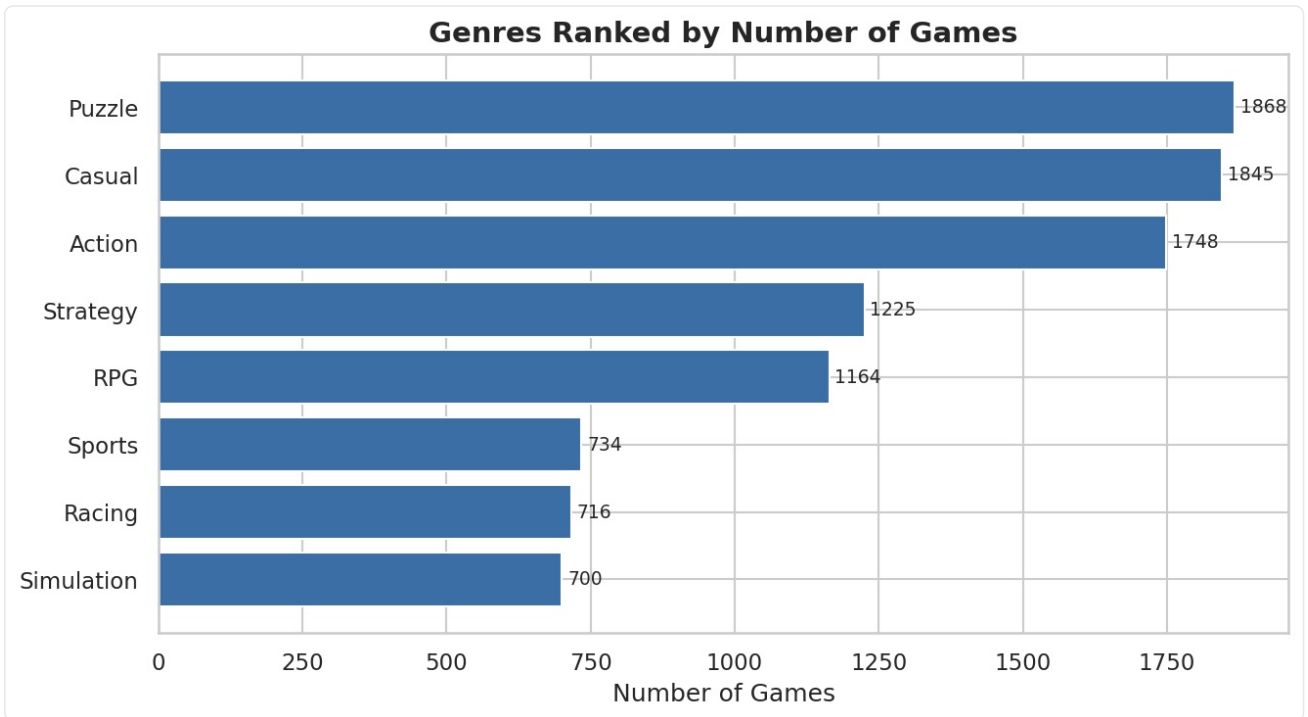
Average rating is almost identical across all four monetization models (3.63–3.66). Freemium games earn the most average reviews (11,483) and downloads (333,167), while Paid games generate by far the highest average revenue per game (\$934,089) — over 9× the

average revenue of Free games (\$97,447).

Monetization Model	Count	Avg. Rating	Avg. Reviews	Avg. Downloads	Avg. Revenue (USD)
Freemium	4,505	3.64	11,483	333,167	\$130,872
Free	3,536	3.64	10,495	314,334	\$97,447
Paid	1,472	3.64	9,727	312,321	\$934,089
Subscription	487	3.66	11,043	274,352	\$875,320

12 Genre Analysis

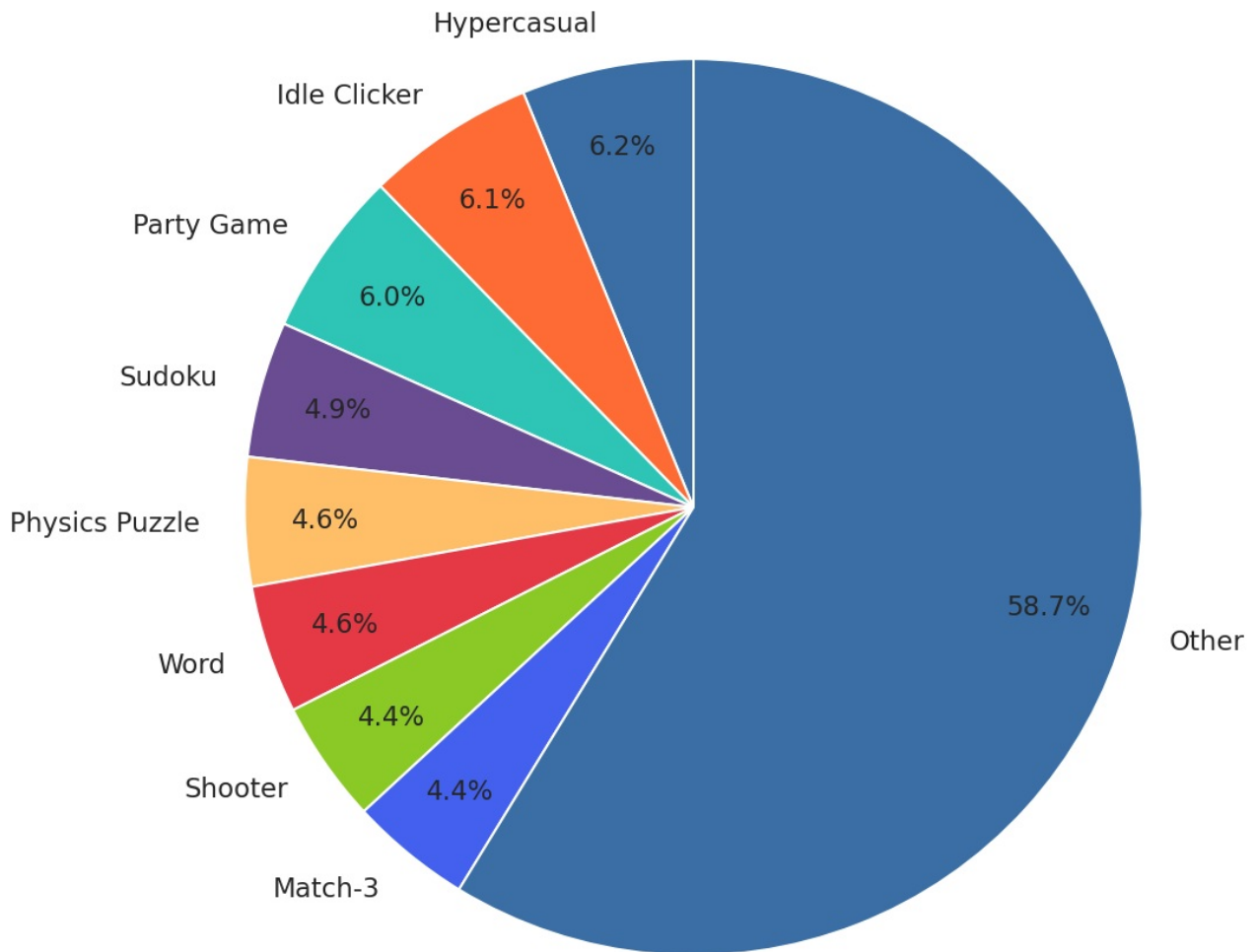
12.1 Genres Ranked



Full ranking of the 8 primary genres by game count, from Racing (716) at the low end to Puzzle (1,868) at the high end.

12.2 Sub-Genre Composition

Sub-Genre Distribution (Top 8 + Other)



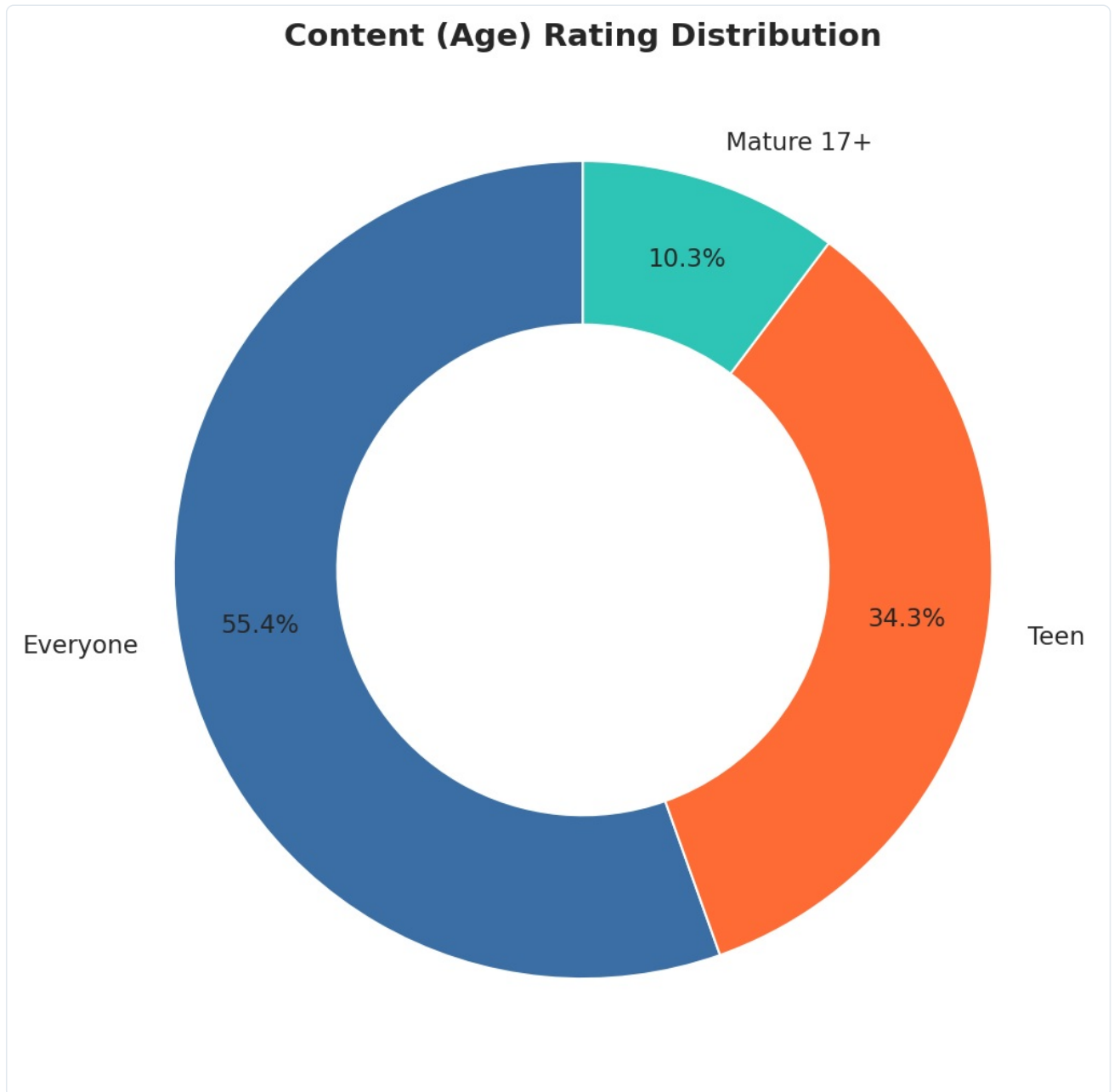
The top 8 sub-genres account for a large share of the catalog, with the remaining 23 sub-genres grouped as "Other."

12.3 Genre Performance Summary

Genre	Count	Avg. Rating	Avg. Downloads	Avg. Reviews
Casual	1,845	3.65	395,582	14,334
RPG	1,164	3.64	384,621	12,961
Action	1,748	3.64	327,286	12,034
Puzzle	1,868	3.65	311,944	9,847
Strategy	1,225	3.63	307,534	8,385
Simulation	700	3.65	247,809	7,744
Sports	734	3.64	242,971	9,061
Racing	716	3.64	202,317	7,306

The most common genre overall is **Puzzle**, but the genre with the highest average downloads per game is **Casual**.

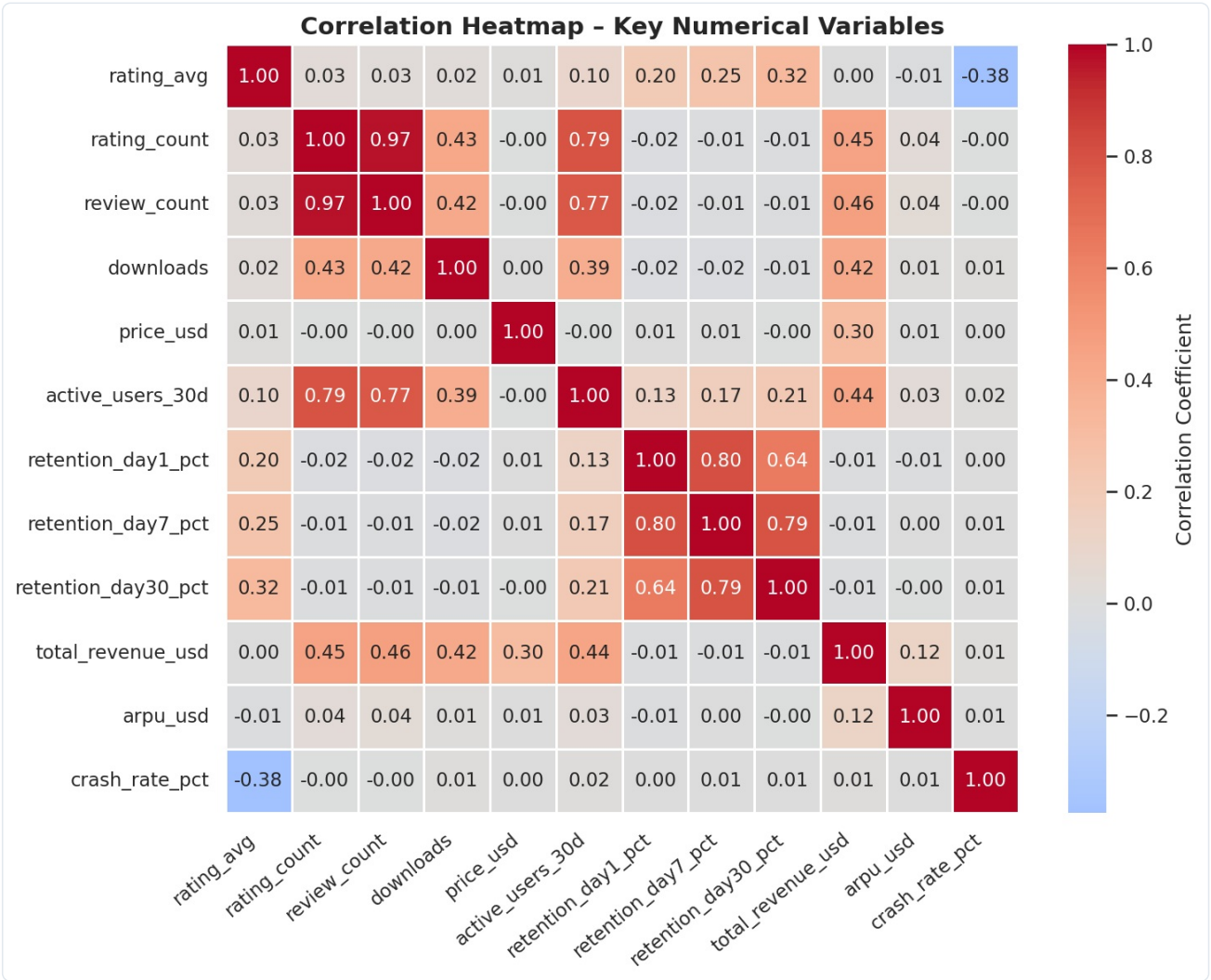
13 Content (Age) Rating Analysis



"Everyone" is the dominant content rating (5,544 games, 55.4%), followed by "Teen" (3,430 games, 34.3%) and "Mature 17+" (1,026 games, 10.3%). This confirms the catalog skews heavily toward broad-audience, family-friendly content.

14 Correlation Analysis

Pearson correlation coefficients were calculated across twelve key numerical variables: rating, rating count, review count, downloads, price, active users, retention (day 1/7/30), total revenue, ARPU, and crash rate.



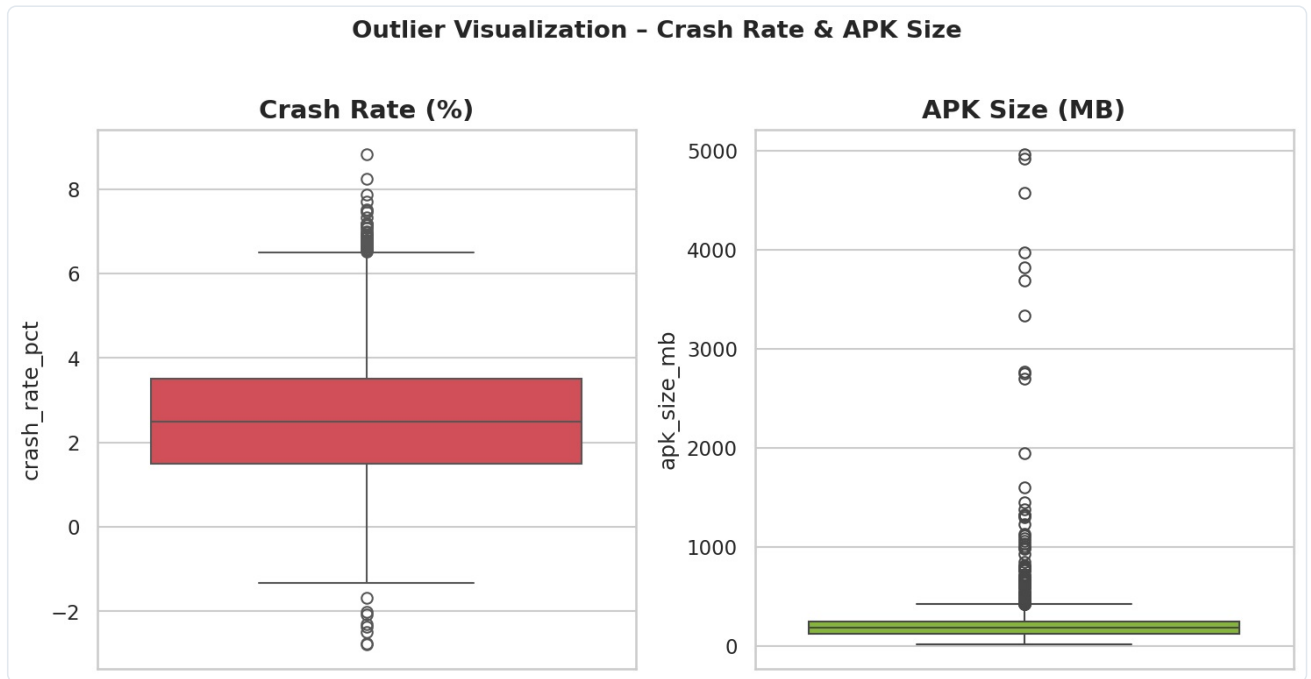
The correlation heatmap condenses all pairwise relationships between the twelve numerical variables into a single visual grid, with dark red indicating strong positive correlation and dark blue indicating negative correlation.

14.1 Strongest Correlations Found

Variable Pair	Correlation (r)	Strength
rating_count ↔ review_count	0.972	Very strong positive
retention_day1_pct ↔ retention_day7_pct	0.805	Strong positive
rating_count ↔ active_users_30d	0.791	Strong positive
retention_day7_pct ↔ retention_day30_pct	0.787	Strong positive
review_count ↔ active_users_30d	0.772	Strong positive
retention_day1_pct ↔ retention_day30_pct	0.639	Moderate positive
review_count ↔ total_revenue_usd	0.464	Moderate positive
rating_avg ↔ review_count	0.032	Negligible

The strongest relationship in the dataset is between `rating_count` and `review_count` ($r = 0.972$), which is expected since both track user engagement volume. Retention at successive checkpoints (Day 1, 7, 30) is also strongly self-correlated, confirming that early retention is a reliable predictor of later retention. Notably, **rating quality (`rating_avg`) shows almost no correlation with popularity metrics** such as review count, download count, or revenue — a highly-downloaded or highly-reviewed game is not necessarily a highly-rated one.

15 Outlier Visualization



Crash rate shows a small number of negative values (invalid, discussed in Section 4.6) and a long tail up to 8.83%. APK size shows several very large outliers up to ~4,967 MB, alongside a typical range of roughly 120–241 MB (IQR).

Not every statistical outlier represents a data error. In this dataset:

- **High-download / high-review outliers** (e.g. the 995 games above the download IQR upper bound) most likely represent genuinely popular, high-performing titles — this is expected in any app marketplace, where a small share of "hit" games capture a disproportionate share of installs and engagement.
- **Negative-value outliers** (crash rate below 0%, price below \$0, rating above 5.0) are logically impossible and should be treated as data-quality issues rather than legitimate extremes (see Section 4.6).
- **Large APK sizes** (multi-gigabyte) are plausible for graphically rich 3D games and are not automatically classified as errors.

16 Key Insights

1. Dataset scale: 10,000 cleaned Android game records across 57 attributes, reduced from 10,150 raw records after removing 150 exact duplicates.

3. Most common sub-genre: Hypercasual (619 games).

5. Most reviewed game: "Game_5704_Rush" (Sports) with 686,581 reviews.

7. Monetization split: Freemium leads at 45.1% of games, followed by Free (35.4%), Paid (14.7%), and Subscription (4.9%).

9. Strongest correlation: rating_count and review_count ($r = 0.972$) — the two engagement-volume metrics move almost in lockstep.

11. Revenue by model: Paid games earn the highest average revenue per title (\$934,089) — over 9× Free games (\$97,447) — despite having far fewer installs.

2. Most common genre: Puzzle (1,868 games, 18.7%), closely followed by Casual (1,845 games).

4. Average rating: 3.64 out of 5, with a tight standard deviation of 0.31.

6. Most downloaded game: "Game_3124_Dash" (RPG) with 53.45 million downloads.

8. Content rating: 55.4% of games are rated "Everyone," 34.3% "Teen," and 10.3% "Mature 17+."

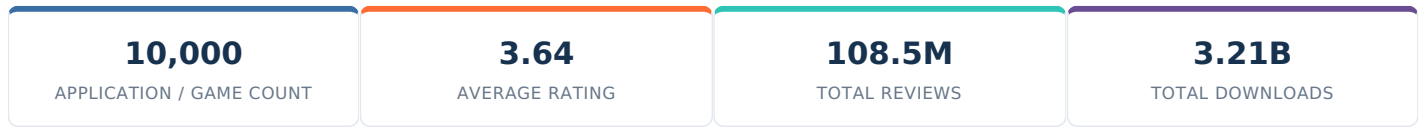
10. Rating independence: rating_avg has near-zero correlation with review_count ($r = 0.032$) and downloads — popularity does not predict quality.

12. Data quality: 10 games have invalid ratings above 5.0 (max 5.98) and 10 have invalid negative prices — flagged for exclusion from range-sensitive statistics.

17 Interactive Dashboard

To make the cleaned dataset explorable beyond static charts, the project includes a design for an interactive HTML/JavaScript/Chart.js dashboard. It surfaces the same KPIs and distributions covered in this report, but allows a user to filter the dataset live and see every chart update in real time.

17.1 Dashboard KPI Cards



17.2 Interactive Filters

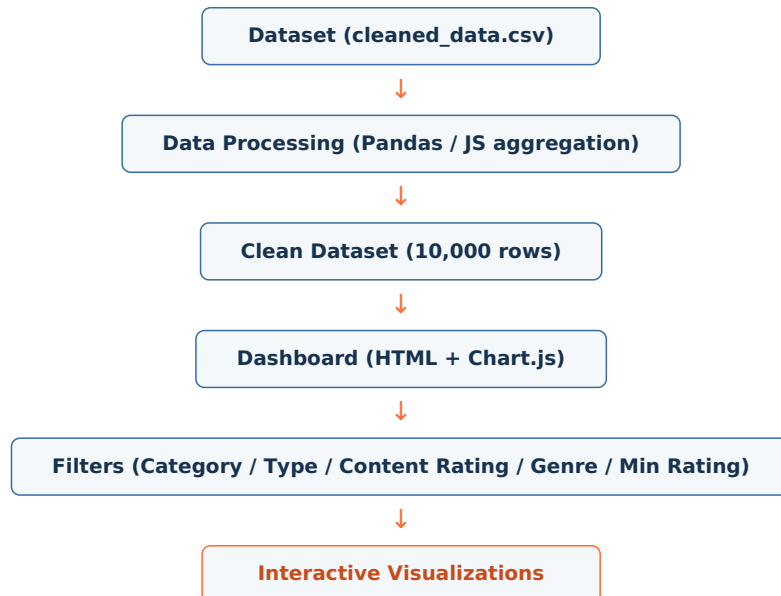


Selecting a filter recomputes the KPI cards and every chart on the page using only the matching subset of the 10,000-game dataset, giving the viewer an on-demand version of the analysis performed in Sections 6–15.

17.3 Dashboard Charts

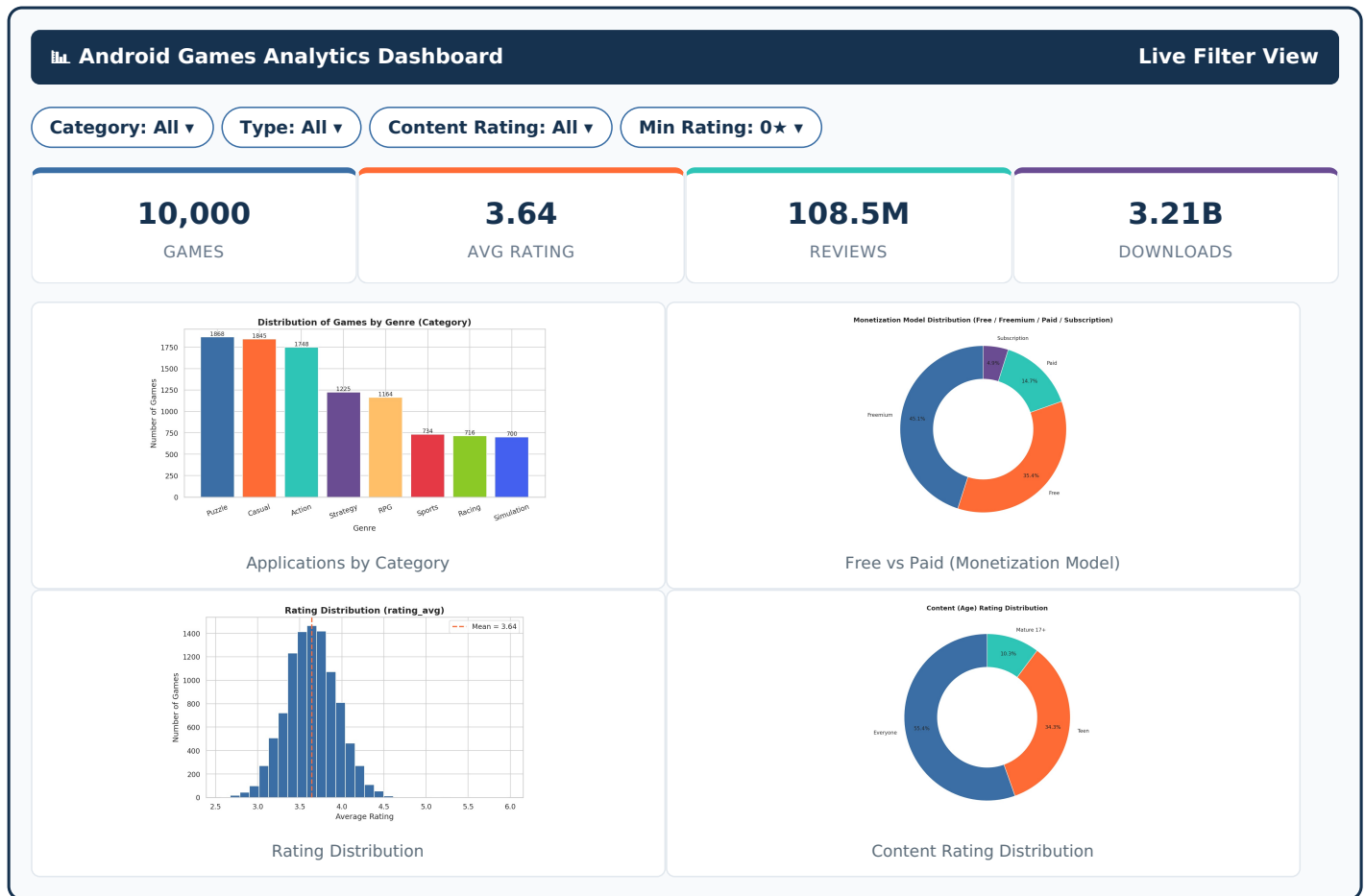
- Applications by Category (genre) — bar chart
- Rating Distribution — histogram
- Monetization Model Split (Free / Freemium / Paid / Subscription) — doughnut chart
- Content Rating Distribution — pie chart
- Genre / Sub-Genre Distribution — bar chart
- Reviews Distribution — histogram
- Rating vs. Reviews — scatter plot

17.4 Dashboard Architecture

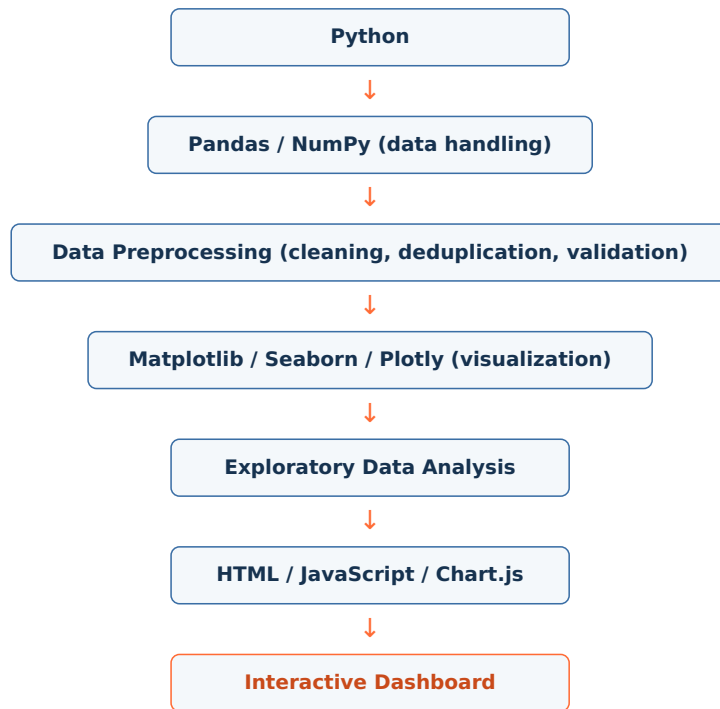


18 Dashboard Visual Representation

The mock-up below illustrates how the KPI cards, filters, and charts come together into a single analytics-dashboard layout, using the actual charts generated from `cleaned_data.csv`.



This dashboard mock-up reuses the real charts produced from the cleaned dataset (Sections 7, 8, 11, and 13) to demonstrate how they would appear when embedded together in a live, filterable dashboard view.



20 Conclusion

This project walked through a full data-analytics pipeline on a 10,000-record Android games dataset — from raw-data inspection through cleaning, exploratory analysis, and dashboard design. Preprocessing proved essential: the raw file contained 150 duplicate records and 20 logically invalid values (ratings above 5.0, negative prices, negative crash rates) that would have skewed averages and correlations if left uncorrected, alongside nine columns with structurally-expected missing data that were preserved as such rather than imputed.

The exploratory analysis revealed a catalog dominated by Puzzle and Casual genres, a tightly-clustered average rating of 3.64, a Freemium-led monetization landscape, and a strong self-correlation between engagement-volume metrics (ratings, reviews, active users) — while showing that rating quality is essentially independent of popularity. Visualization made these patterns immediately legible, and the proposed interactive dashboard extends that legibility into an explorable, filterable tool.

Overall, the project demonstrates that rigorous preprocessing, disciplined EDA, and thoughtful visualization together turn a messy, real-world-style dataset into a trustworthy foundation for decision-making and further analysis.

21 Future Enhancements

- **Machine learning rating prediction:** Train a regression model (e.g. Random Forest, XGBoost) to predict `rating_avg` from genre, monetization model, retention, and engagement features.
- **Popularity / "hit game" prediction:** Use the existing `is_hit_game` flag as a target for a classification model that predicts breakout success from early-stage signals such as pre-launch wishlist count and Day-1 retention.
- **Recommendation system:** Build a content-based or collaborative filtering engine using genre, sub-genre, and art-style similarity.
- **Revenue forecasting:** Model `total_revenue_usd` and `post_30d_revenue_usd` as time-series or regression targets using marketing spend, CPI, and conversion rate features.
- **Deployment as a live web application:** Convert the dashboard mock-up (Section 18) into a deployed Chart.js + Flask/Node application with real filter interactivity.
- **Real-time data updates:** Connect the dashboard to a live data pipeline so KPIs refresh as new game records arrive.
- **Advanced filtering:** Add multi-select filters, date-range filters (release date), and revenue-band filters.
- **Cloud deployment:** Host the dashboard and underlying data pipeline on a cloud platform (e.g. AWS, GCP, Azure) for scalable public access.

Report generated from the actual contents of `android_games_eda_ready.csv` (raw, 10,150 rows) and `cleaned_data.csv` (cleaned, 10,000 rows). All statistics, charts, and findings in this report are computed directly from these files using Python (Pandas, NumPy, Matplotlib, Seaborn) — no values were fabricated or estimated.